

# REPORT ON INDUSTRIAL ATTACHMENT



CARRIED OUT AT

**NZOIA SUGAR COMPANY (2025) LIMITED**

LABORATORY DEPARTMENT

BY

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REGISTRATION NUMBER: SCP213-1123/2023

A report submitted to the Department of Chemistry, Jomo Kenyatta University of Agriculture and Technology (JKUAT), in partial fulfilment of the requirements for the industrial attachment component of the Bachelor of Science in Analytical Chemistry.

**JULY - AUGUST 2026**

# DECLARATION

## **Student declaration**

I declare that this report is my own original work and has not been presented for any award in any other institution.

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## **Company supervisor's declaration**

This report has been submitted for review with my approval as the company supervisor.

Name: Mr. Boniface Ouma

Signature: \_\_\_\_\_

Date: \_\_\_\_\_

## ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the Almighty God for granting me the strength and opportunity to undertake this industrial attachment successfully.

My heartfelt appreciation goes to my company supervisor, Mr. Boniface Ouma (Processing Manager) and his junior team (Assistant Processing Manager, Lab Chemist, assistant Lab Chemist), for their patient mentorship, willingness to explain procedures, and for giving me hands-on exposure in the Quality Control laboratory.

I extend my thanks to the entire Laboratory team at Nzoia Sugar Company for welcoming me, sharing their knowledge, and allowing me to participate actively in daily analytical work. Their cooperation made this experience both educational and enjoyable.

I also wish to thank the Department of Chemistry at Jomo Kenyatta University of Agriculture and Technology (JKUAT) for organizing and facilitating this attachment as part of my Bachelor of Science in Analytical Chemistry program.

Finally, I am grateful to my family and friends for their encouragement and support throughout this period.

## ABSTRACT

This report presents the findings and experience gained during an industrial attachment undertaken at Nzoia Sugar Company Limited, in the Laboratory department, between July and August 2026. The attachment was carried out in partial fulfilment of the requirements for the Bachelor of Science in Analytical Chemistry degree at Jomo Kenyatta University of Agriculture and Technology (JKUAT).

During the attachment period, hands-on exposure was gained in a wide range of analytical procedures used to monitor sugar production quality and process efficiency, including determination of Brix, Pol, and purity across juices, syrups, massecuites, and molasses grades; reducing sugar and sucrose analysis; ash and moisture content; pH and colour determination; fibre and preparation index testing on cane; boiler water quality analysis; and effluent monitoring through Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) testing.

Competence was developed in the use of key laboratory instruments, including the polarimeter, refractometer, drying oven, muffle furnace, spectrophotometer, and titration apparatus.

The attachment strengthened practical laboratory skills such as accurate sample handling, adherence to standard operating procedures, and data recording, while reinforcing the theoretical foundations covered in university coursework. It also provided insight into the role of quality control in ensuring compliance with KEBS, ICUMSA, and NEMA regulatory standards.

Overall, the attachment achieved its stated objectives, providing a valuable bridge between classroom theory and industrial practice, and strengthening readiness for both further academic work and future employment in analytical or quality control roles.

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## LIST OF ABBREVIATIONS

| Abbreviations    | Meaning                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| AH               | A-Heavy Molasses (mother liquor separated from A-massecuite)                                                                                     |
| AL               | A-Light Molasses / A-Liquor (rich washings from A-sugar centrifugation)                                                                          |
| BOD <sub>5</sub> | Biochemical Oxygen Demand (5-day incubation at 20 °C)                                                                                            |
| BFW              | Boiler Feed Water                                                                                                                                |
| BOD              | Biochemical Oxygen Demand                                                                                                                        |
| BH               | B-Heavy Molasses (mother liquor separated from B-massecuite)                                                                                     |
| Brix (°Bx)       | Percentage of Total Dissolved Solids by Mass                                                                                                     |
| CJ               | Clear Juice                                                                                                                                      |
| COD              | Chemical Oxygen Demand                                                                                                                           |
| DO               | Dissolved Oxygen                                                                                                                                 |
| DW               | Distilled Water                                                                                                                                  |
| EBT / Solochrome | Solochrome Black T / Eriochrome Black T Complexometric Indicator                                                                                 |
| EDTA             | Ethylenediaminetetraacetic Acid Disodium Salt (C <sub>10</sub> H <sub>14</sub> N <sub>2</sub> Na <sub>2</sub> O <sub>8</sub> ·2H <sub>2</sub> O) |

|                                          |                                                                                                |
|------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>ETP</b>                               | Effluent Treatment Plant                                                                       |
| <b>FAS</b>                               | Ferrous Ammonium Sulfate $((\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O})$ |
| <b>FC</b>                                | Filter Cake (Filter Mud)                                                                       |
| <b>FEJ</b>                               | First Expressed Juice                                                                          |
| <b>FJ</b>                                | Filter Juice (Rotary Drum Vacuum Filtrate)                                                     |
| <b>GCV</b>                               | Gross Calorific Value                                                                          |
| <b>GLP</b>                               | Good Laboratory Practice                                                                       |
| <b>ICUMSA</b>                            | International Commission for Uniform Methods of Sugar Analysis                                 |
| <b>IU</b>                                | ICUMSA Units                                                                                   |
| <b>JKUAT</b>                             | Jomo Kenyatta University of Agriculture and Technology                                         |
| <b>LEJ</b>                               | Last Expressed Juice                                                                           |
| <b>M-Alkalinity</b>                      | Methyl Orange Alkalinity (Total Alkalinity)                                                    |
| <b>MASSEC</b>                            | Masseccuite (Dense mixture of sugar crystals suspended in mother liquor)                       |
| <b>ME</b>                                | Milling Extraction Efficiency                                                                  |
| <b>MJ</b>                                | Mixed Juice                                                                                    |
| <b>MGSF</b>                              | Multi-Grade Sand Filter                                                                        |
| <b>NEMA</b>                              | National Environment Management Authority (Kenya)                                              |
| <b>NTU</b>                               | Nephelometric Turbidity Units                                                                  |
| <b>OxyTop</b>                            | Closed-System Respirometric Pressure Sensor System for BOD Measurement                         |
| <b>P-Alkalinity</b>                      | Phenolphthalein Alkalinity                                                                     |
| <b>PAC</b>                               | Poly-Aluminum Chloride $([\text{Al}_2(\text{OH})_n\text{Cl}_{6-n}]_m)$                         |
| <b>PAM</b>                               | Polyacrylamide Polymer Flocculant                                                              |
| <b>Pb(CH<sub>3</sub>COO)<sub>2</sub></b> | Lead Sub acetate / Lead Acetate Clarifying Agent                                               |
| <b>PI</b>                                | Preparation Index                                                                              |
| <b>Pol</b>                               | Polarization / Polarimetric Sucrose Content                                                    |
| <b>POL_B</b>                             | Percentage Pol (Sucrose Content) in Bagasse                                                    |
| <b>POL_FC</b>                            | Percentage Pol (Sucrose Content) in Filter Cake                                                |

|                   |                                                                                |
|-------------------|--------------------------------------------------------------------------------|
| <b>PS-DVB</b>     | Polystyrene-Divinylbenzene Matrix                                              |
| <b>PPE</b>        | Personal Protective Equipment                                                  |
| <b>RS</b>         | Reducing Sugars (Glucose and Fructose)                                         |
| <b>SAC</b>        | Strong Acid Cation Exchange Resin                                              |
| <b>TDS</b>        | Total Dissolved Solids                                                         |
| <b>TH</b>         | Total Hardness (expressed as mg/L CaCO <sub>3</sub> )                          |
| <b>TRS</b>        | Total Reducing Sugars                                                          |
| <b>TSS</b>        | Total Suspended Solids                                                         |
| <b>UB</b>         | Unopened Cells / Unbroken Cell Structure                                       |
| <b>UV-Vis</b>     | Ultraviolet-Visible Spectrophotometry                                          |
| <b>VFA</b>        | Volatile Fatty Acids                                                           |
| <b>WTP</b>        | Water Treatment Plant                                                          |
| <b>°Z</b>         | Degrees Zuccherò (International Sugar Scale Unit on Saccharimeters)            |
| <b>α-Naphthol</b> | 1-Naphthol Qualitative Colorimetric Reagent (C <sub>10</sub> H <sub>8</sub> O) |

# CHAPTER 1: INTRODUCTION

## 1.1 BACKGROUND OF INDUSTRIAL ATTACHMENT

Industrial attachment is a mandatory component of the Bachelor of Science in Analytical Chemistry programme at Jomo Kenyatta University of Agriculture and Technology (JKUAT), designed to bridge the gap between classroom theory and real-world industrial practice. While university coursework equips students with theoretical knowledge of analytical techniques, instrumentation, and chemical principles, industrial attachment exposes students to how this knowledge is applied in an actual production and quality control environment.

This report documents the industrial attachment undertaken at Nzoia Sugar Company Limited, in the Laboratory department, over a period running through July and August 2026. During this period, the attachment provided hands-on exposure to routine analytical procedures used in monitoring sugar production quality and factory effluent, under the supervision of Mr. Boniface Ouma (company supervisor).

## 1.2 OBJECTIVES

### 1.2.1 General Objective

To develop the practical laboratory skills and industry exposure necessary to competently perform quality control analysis in a real manufacturing environment, in fulfilment of the industrial attachment requirement for the Bachelor of Science in Analytical Chemistry programme.

### 1.2.2 Specific Objectives

1. To understand the organizational structure and operations of Nzoia Sugar Company, with particular focus on the Quality control/Laboratory department.
2. To perform and understand routine analytical tests used in sugar quality control, including Brix, Pol, purity, ash content, moisture, colour, and pH determination.
3. To gain competence in the use of laboratory instruments such as the polarimeter, refractometer, drying oven, OxiTop apparatus, and COD reactor.
4. To understand the role of quality control testing in ensuring product compliance with KEBS and ICUMSA standards.
5. To develop practical laboratory skills, including sample handling, safety practices, and data recording/reporting.
6. To identify challenges encountered in real industrial analytical work and how they are addressed in practice.

## 1.3 SCOPE AND LIMITATIONS

This report covers activities undertaken within the Quality Control/Laboratory department at Nzoia Sugar Company over the attachment period (July–August 2026)

Includes:

- **Daily sugar material analysis:** Sugar Pol, Sugar Moisture, Sugar Colour, Water Insoluble Matter in Sugar, and Crystal Size of Sugar
- **Weekly special analysis rotation, including:**
  - **Titration Acidity** in FEJ, MJ, CJ, and Syrup (Monday and Friday)
  - Dextran and Starch in FEJ, MJ, CJ, and Syrup (Monday and Friday)
  - **Reducing Sugar and Total Reducing Sugar** in Final Molasses (Tuesday and Saturday)
  - **Preparation Index and Sucrose Analysis** (Wednesday)
  - **Colour** of CJ, Syrup, AL, and Melt, and Fibre % Cane (Thursday)
  - **Quality checks on process chemicals:** CaO in Lime, Strength of Caustic Soda, and Settling Test of Juice Flocculant
  - Weekly calibration of laboratory equipment
  - **Cane quality analysis**, including fibre content, preparation index, and Pol % cane
  - **Boiler water quality analysis** (hardness, alkalinity, silica, sulphite, TDS, chloride)
  - **Effluent and environmental testing**, including BOD and COD determination
  - General laboratory practices observed and applied, including sample collection, calibration, and safety procedures

This report does not cover in depth the operations of other departments such as Agriculture, Engineering, or Finance, except where relevant background is needed to understand the company's overall operations.

## CHAPTER 2: COMPANY PROFILE

### 2.1 HISTORY OF NZOIA SUGAR COMPANY

Nzoia Sugar Company Limited (NSC) was established in 1975 under the Companies Act Cap. 486 of the Laws of Kenya, and was issued a certificate of incorporation on 1st August 1975 following an agreement between the Governments of Kenya and France. It was the sixth sugar project to be launched in Kenya, aimed at supporting the government's policy of self-sufficiency in sugar production. Site works began in 1976, factory construction started in 1977, and the first sugar deliveries were made in July 1978.

The company is located in Bungoma County, about 5 kilometres from Bukembe off the Webuye-Bungoma highway. It is a state-owned enterprise, with the Government of Kenya holding the majority shareholding. Over the decades, management of the factory has changed hands several times between the Government and private management contractors. In 2025, following years of reduced capacity, Nzoia Sugar was leased to West Kenya Sugar Company under a 30-year arrangement as part of wider reforms in Kenya's state-owned sugar sector, and resumed milling after rehabilitation of critical plant infrastructure.

### 2.2 VISION, MISSION AND CORE VALUES

#### Vision

To be a leading, efficient, and competitive producer of high-quality sugar and associated products in East Africa.

#### Mission

To sustainably produce high-quality sugar and co-products efficiently and profitably through modern technology and skilled human resources, while satisfying customer needs and empowering community development.

#### Core Values

- **Integrity:** Maintaining transparency, honesty, and accountability in all operations.
- **Quality:** Adhering to strict product standards and process metrics.

- **Innovation:** Utilizing modern technology and research for efficiency.
- **Teamwork:** Fostering collaboration across operational departments.
- **Environmental Responsibility:** Commitment to ecological preservation and statutory compliance.

## 2.3 ORGANIZATIONAL STRUCTURE

### Executive Governance & Oversight

The governance framework of Nzoia Sugar Company Limited is anchored on corporate compliance, risk management, and strategic oversight. The apex authority flows as follows:

- **Board of Directors**

Formulates long-term corporate policies, approves strategic plans, and ensures governance compliance

- **Managing Director & Chief Executive Officer (MD / CEO)**

Executes Board directives and drives day-to-day operations across all directorates.

- **Internal Audit & Risk Management**

Reports directly to the Board Audit Committee to guarantee financial and operational transparency.

- **Legal Services & Corporation Secretariat**

Manages institutional statutory compliance, legal risks, and board transactions.

### Core Operational Directorates and Departments

The institutional hierarchy divides operations into core revenue-generating wings and strategic support units:

#### Agriculture Department

- Responsible for nucleus estate management, cane development, land preparation, agronomy, and public sugar research.
- Oversees outreach programs, extension services for outgrower farmers, and heavy machinery fleet management.

#### Engineering & Factory Process Operations Departments



- Manages factory plant milling infrastructure, instrumentation, process control, and preventative maintenance scheduling.
- Monitors power generation, steam generation, and factory chemical processing lines.

### **Finance & Accounts Department**

- Controls corporate budgeting, financial reporting, management accounting, and tax compliance.
- Houses sections for payroll operations, general ledger, accounts payable, accounts receivable, and outgrower payment handling. Procurement & Supply Chain Management
- Governs public procurement procedures, raw material sourcing, inventory control, and central warehousing. Human Resources & Administration
- Directs talent acquisition, industrial relations, staff training, corporate health, safety, and physical plant security. Sales & Marketing Department.

# NZOIA SUGAR COMPANY LIMITED ORGANIZATIONAL CHART

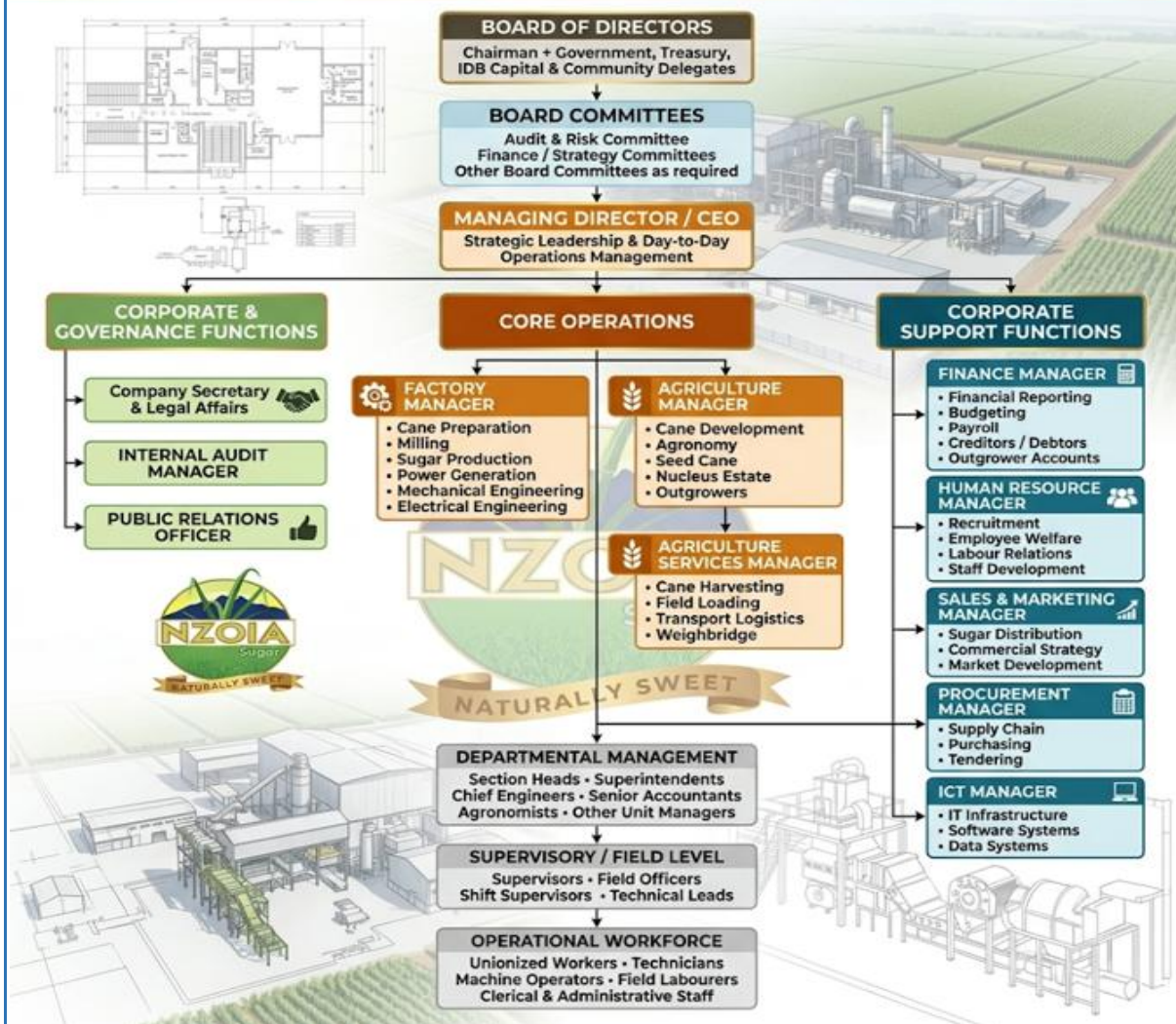


Figure 1: Organization Structure

# CHAPTER 3: LITERATURE REVIEW

## 3.1 OVERVIEW OF SUGAR MANUFACTURING CHEMISTRY

Sugarcane is composed mainly of sucrose, water, fibre, and smaller amounts of reducing sugars, minerals, and organic non-sugars. The manufacturing process begins with milling/extraction, where cane is crushed to release juice. The raw juice is then clarified using heat and lime to remove impurities, concentrated by evaporation into syrup, and finally crystallized under vacuum to separate sugar crystals from the remaining molasses. The crystals are then separated by centrifugation, dried, graded, and packaged.

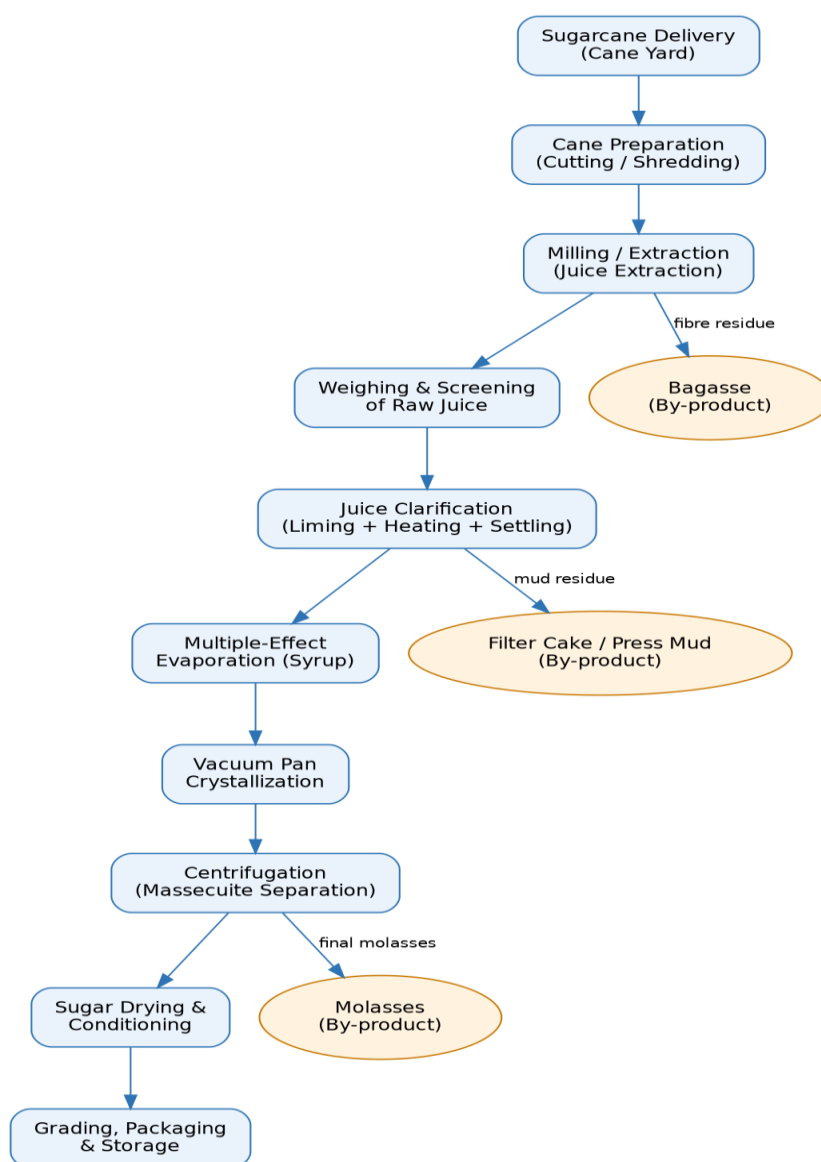


Figure 2: Overall Sugar Manufacturing Process Flow, from cane delivery to packaged sugar, with by-product streams (bagasse, filter cake, molasses) indicated.

### **3.2 ROLE OF THE QUALITY CONTROL LABORATORY**

The QC laboratory monitors product quality at every stage of production, from incoming cane to the final packaged sugar. Regular testing ensures the process is running efficiently (e.g. checking that as much sucrose as possible is being extracted and recovered), that the final product meets regulatory and customer specifications, and that any deviations are detected and corrected early. Without consistent QC testing, a company risks producing off-specification sugar, financial losses from poor extraction efficiency, and non-compliance with food safety regulations.

### **3.3 QUALITY STANDARDS AND REGULATORY BODIES**

In Kenya, the Kenya Bureau of Standards (KEBS) sets and enforces national specifications that sugar and related products such as molasses must meet before sale, covering parameters like Brix, sulphated ash, pH, and microbial limits. KEBS testing also draws on internationally recognized methods published by the International Commission for Uniform Methods of Sugar Analysis (ICUMSA), founded in 1897, which provides standardized laboratory procedures for analyzing raw, cane, white, and specialty sugars so that results from different laboratories and countries can be compared reliably.

### **3.4 ANALYTICAL TECHNIQUES USED IN SUGAR QC**

Several core analytical techniques underpin sugar QC work. Titration is used for tests such as sulphite and reducing sugar determination. Polarimetry measures the optical rotation of a sugar solution to determine Pol (apparent sucrose content), exploiting the fact that sucrose rotates plane-polarized light. Refractometry measures the refractive index of a solution to quickly estimate Brix (total dissolved solids). Spectrophotometry is used for colour determination in ICUMSA units by measuring light absorbance at a specified wavelength.

### **3.5 ENVIRONMENTAL REGULATIONS AND EFFLUENT STANDARDS**

Beyond product quality, sugar manufacturing generates process effluent that must be treated and monitored before discharge, making environmental compliance an additional responsibility of the QC laboratory. In Kenya, effluent discharge is regulated under the Environmental Management and Co-ordination Act (EMCA) of 1999, which empowers the National Environment Management Authority (NEMA) to set and enforce discharge standards and to license effluent discharge into the environment.

Under NEMA's Water Quality Regulations, treated effluent released into water bodies or onto land is required to meet defined maximum limits for key parameters, including a 5-day Biochemical Oxygen Demand (BOD<sub>5</sub>) of not more than 30 mg/L, a Chemical Oxygen Demand (COD) of not more than 50 mg/L, and a pH within the range of 6.5 to 8.5, among other physical, chemical, and biological parameters.

These limits give the BOD and COD tests performed in the lab a direct regulatory purpose: results are compared against the NEMA thresholds to confirm that the factory's effluent treatment plant is functioning adequately before water is released or reused, and persistent exceedances would expose the company to potential penalties or licence complications. This places environmental testing alongside product-quality testing as a core function of the laboratory, rather than a secondary activity.

### **3.6 QUALITY ASSURANCE VERSUS QUALITY CONTROL**

Although often used interchangeably, Quality Assurance (QA) and Quality Control (QC) refer to distinct but complementary aspects of a quality management system.

Quality Assurance is the broader, proactive set of planned activities aimed at preventing defects before they occur — it covers how processes, procedures, and standards are designed and followed, including staff training, standard operating procedures (SOPs), instrument calibration schedules, and documentation practices.

Quality Control, by contrast, is the reactive, product-focused set of activities used to verify that outputs meet specified requirements — it involves the actual testing and inspection of samples, such as the Brix, Pol, ash, and effluent tests described in this report, to detect defects or deviations after they have occurred.

In practical terms, QA is concerned with "are we doing things the right way?" while QC is concerned with "is this specific output right?" The two functions depend on each other: QC testing data (such as recurring out-of-specification results) often feeds back into QA processes to correct or improve procedures, while a strong QA framework (properly calibrated instruments, trained staff, validated methods) is what makes individual QC test results reliable in the first place.

Most of the day-to-day work covered in Chapter 5 of this report falls under Quality Control, while the standard procedures, KEBS/ICUMSA method compliance, and instrument calibration practices described in Chapters 3 and 4 reflect the underlying Quality Assurance framework that supports it.

## CHAPTER 4: SUGAR PROCESSING

### 4.1 Cane Quality Control, Raw Material Valuation, and Mass Quantification

#### 3.1.1 Incoming Materials Inspection and Analytical Sampling

The reception bay serves as the primary technical gatehouse where all incoming raw material undergoes strict quality auditing, sampling, and mass determination.

Monitoring sugarcane quality is critical, as it directly impacts processing efficiencies, factory extraction yields, clarification chemistry, and the final colour index of the manufactured sugar crystals.

Physical inspection evaluates the cleanliness and composition of each consignment against established operational thresholds:

- **Extraneous Matter Assessment:** Deliveries are systematically audited to quantify non-cane impurities, including dry leaves (trash), immature cane tops, mud, rock fragments, and root clusters. Plant engineering specifications require total extraneous matter to remain strictly under **8.0% by total weight**. Consignments exceeding this limit drive up the mill's fiber loading, mechanically trap and absorb sucrose, and degrade overall milling extraction efficiency.
- **Logistical Traceability:** Deliveries are logged by geographic origin to track performance across internal Nucleus estates, local Outgrower farms (via short-haul tractor-trailers), and distant outstations (including Makuti, Misikhu, Busia, Chemilil, Miwani, Kericho, Naitiri, and Khalaba) utilizing long-haul heavy commercial trucks.
- **Juice Sampling Protocols:** Laboratory technicians execute randomized sampling routines across incoming transport beds using two core methodologies:
  - **Spot/Systematic Sampling:** A single, undamaged, full-length cane stalk is extracted from a designated section of the truck bed to undergo immediate physical profiling and juice chemistry verification.
  - **Composite Sampling:** Incremental core samples are drawn from various depths and locations within a bulk delivery at scheduled intervals. These are combined into a single, homogeneous sample to represent the average quality profile of the entire delivery.

#### 4.1.2 Cane Maturity Modeling and Age Verification

Accurately determining sugarcane maturity is vital to prevent processing immature or over-mature crops, both of which suffer from depressed sucrose concentrations. Technicians verify crop age in both field surveys and receiving labs by conducting physical internode counts on representative stalks:

$$\text{Sugar age (Months)} = (X + 6) \times 2/3$$

### What This Formula Represents

In the context of sugarcane agronomy (such as operations at Nzoia Sugar Company), this type of formula is used by field officers to estimate sugarcane maturity and plan harvesting schedules based on a field parameter X.

Where:

1. **X (The Variable):** This typically represents a field indicator, such as the number of visible internodes (nodes) on a sample cane stalk.
2. **The Constant (+6):** This accounts for the initial germination and tillering phase (usually taking 4 to 6 months) before active stalk elongation and rapid node formation begin.
3. **The Fraction (times 2/3):** This is a biological growth-rate constant. It adjusts the physical stalk measurements back into a timeline of months, factoring in regional growth speeds.

### Example Calculation

If a field officer samples a cane crop and determines that  $X = 18$ :

$$\text{Age} = (18 + 6) \times 2/3 = 24 \times 2/3 = 16 \text{ Months}$$

In Western Kenya sugar zones, soft mill cane is typically managed around these calculated timelines to forecast the ideal sucrose-harvesting window (usually between 14 to 18 months).

### 4.1.3 Crop Classification, Varietal Profiling, and Post-Harvest Degradation

Sugarcane entering the processing stream is cataloged by its specific hybrid variety and classified by its physical state and post-harvest age.

### Agronomic Varietal Distribution

The agricultural extension team promotes the cultivation of specific high-sucrose, high-tonnage, and disease-resistant hybrids optimized for the local soil profiles and rainfall patterns:

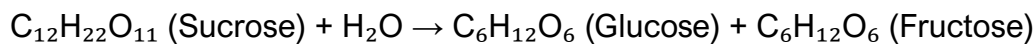
- **Co 421 & Co 420:** High-sucrose, medium-maturity commercial mainstays that provide consistent juice yields.
- **Co 945:** A high-fiber, deep-rooting variety selected for its high drought tolerance.

- **EAK and D8484:** Rapid-maturing, early-season hybrids designed to achieve peak sucrose content within a shortened **14–16-month** window.

### Post-Harvest Physical Classification

- **Fresh Green Cane:** Sugarcane harvested and delivered to the factory gate within a **24-hour** window. This material exhibits peak juice purity >80% and suffers negligible sucrose degradation.
- **Burnt Cane:** Crops from fields intentionally burned prior to cutting to clear out dense undergrowth and trash. Burnt cane demands rapid processing within 24 hours of firing. Delays trigger the growth of *Leuconostoc* bacteria, leading to dextran synthesis and organic acid formation. These compounds increase syrup viscosity, causing the juice to stick to heat-exchanger tubes and foul boiling surfaces.
- **Stale Cane:** Unburnt or burnt cane that has been delayed in transit or left in storage yards for **longer than 24 hours** post-harvest. This delay initiates severe acid hydrolysis—commonly known as **sucrose inversion**. During inversion, plant invertase enzymes and bacterial activity break down crystallizable sucrose into non-crystallizable reducing sugars (glucose and fructose).

### Sucrose Inversion Chemistry:



*Catalyst: Invertase / Acid*

### Standard Juice Laboratory Assays

1. **Brix Value (°Bx):** The percentage of total dissolved solids by mass in the juice, quantified via an optical refractometer.
2. **Polarization (Pol %):** The apparent sucrose concentration determined by measuring the optical rotation of polarized light through a clarified juice sample. The target maturity profile for incoming cane is a Pol value of approximately 14.5%.
3. **Purity Index (%):** A key efficiency metric calculating the ratio of true sugar to total dissolved solids:  

$$\text{Purity\%} = \text{Pol/Brix} \times 100$$

#### 4.1.4 Weighbridge Engineering and Mass Balance Calculations

The industrial weighbridge station utilizes electronic load-cell systems to capture highly accurate mass data. This data is essential for managing outgrower financial payouts, cross-docking transport logistics, and calculating the factory's internal mass balances.

### Weight Vectors and Commercial Deductions



- **Gross Weight:** The total combined mass of the commercial transport vehicle and its full payload of sugarcane upon arrival.
- **Tare Weight:** The mass of the empty transport vehicle logged upon its departure through the exit gates.

**Net Cane Weight:** The true processable biomass weight calculated by adjusting the raw mass values for recorded extraneous matter and trash percentages:

$$\text{Net Weight (kg)} = (\text{Gross Weight} - \text{Tare Weight}) - 160 \text{ kg}$$

The **160 kg fixed deduction** accounts for non-cane extraneous matter, including dirt, mud, trash, and roots.

#### 4.1.5 Weighbridge Strain-Gauge Load Cells

##### 1. Core Working Principle

Weighbridge load cells convert mechanical force (vehicle weight) into a measurable electrical signal through a four-part process:

- **Mechanical Deflection:** The weight of a vehicle causes a steel block (spring element) to microscopically bend.
- **Resistance Change:** Zigzag metallic foil patterns (strain gauges) glued to the block stretch or compress, altering their electrical resistance.
- **Circuit Balance:** Four gauges form a **Wheatstone Bridge** circuit. Unbalancing this circuit generates a proportional output voltage measured in millivolts (mV).
- **Signal Aggregation:** A centralized junction box sums up the individual voltages from all load cells to display an accurate, uniform weight measurement.

##### 2. Mathematical Formula

The voltage output (V-out) is governed by the Wheatstone Bridge equation:

$$V_{\text{out}} = V_{\text{exc}} \times \left\{ R_3 / (R_3 + R_4) - R_2 / (R_1 + R_3) \right\}$$

- **At Rest:** All resistances R are equal, resulting in  $V_{\text{out}} = 0 \text{ mV}$ .
- **Under Load:** The output scales linearly with mechanical strain  $\epsilon$  and the Gauge Factor GF:

$$V_{\text{out}} = V_{\text{exc}} \times \text{GF} \times \epsilon$$

##### 3. Technology Comparison

- **Analog Load Cells:** Simpler and cheaper, but vulnerable to electrical noise, signal tampering, weather interference, and require manual multimeter troubleshooting.

- **Digital Load Cells:** Process data using an internal microprocessor. They offer built-in encryption against fraud, self-diagnostic fault detection, and total immunity to signal interference.

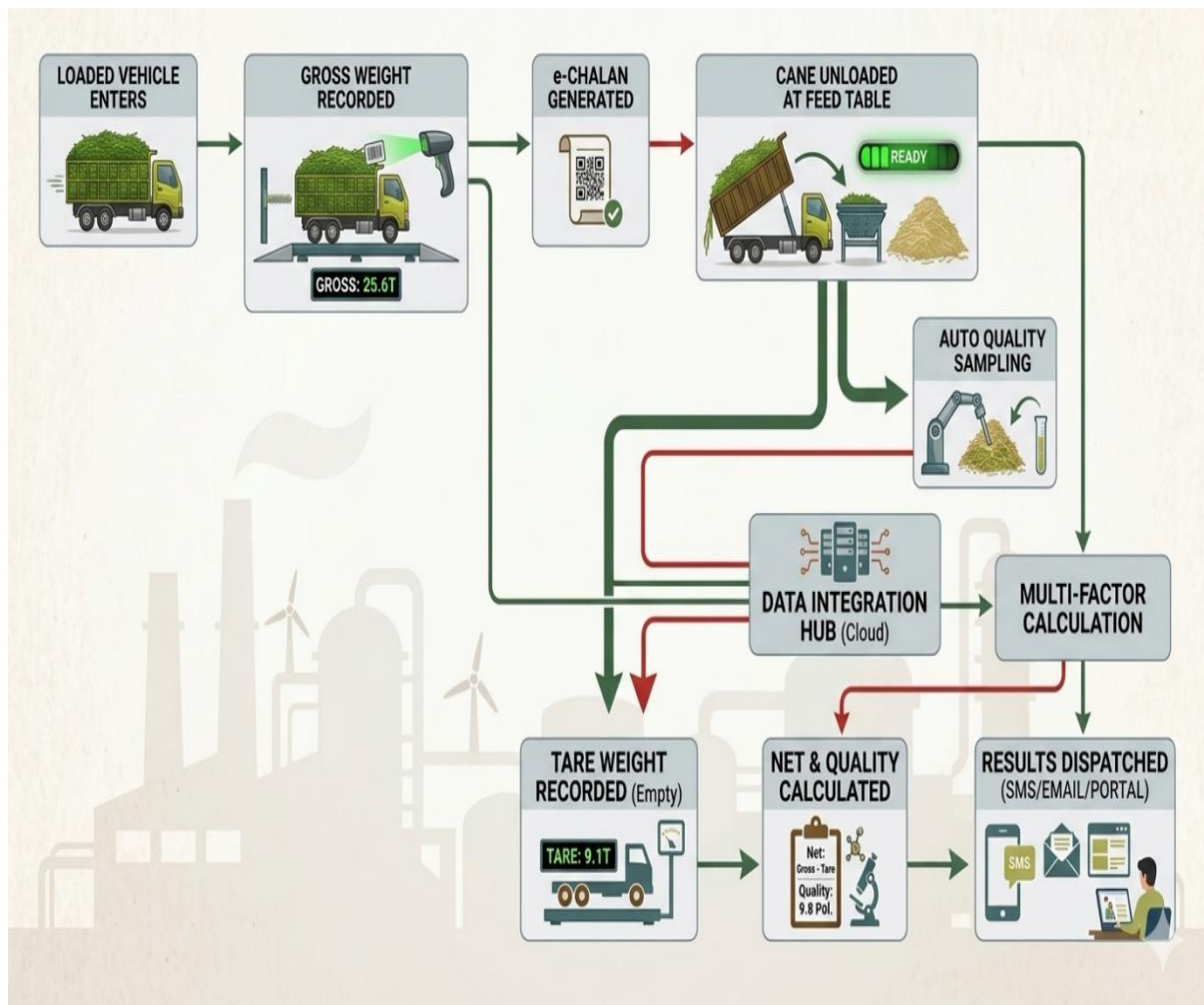


Figure 3: Weighbridge Strain-Gauge Load Cells

## 4.2 Industrial Juice Extraction and Purification Operations

### 4.2.1 Upstream Mechanical Disintegration and Cane Condition Monitoring

Raw sugarcane delivered to the feeding tables undergoes intensive mechanical conditioning before fluid extraction. Crushing intact stalks under compression rollers is highly inefficient; therefore, preparing the stalks maximizes accessible surface area and open cell parameters.

1. **Primary Knife Pass (Levelling System):** High-speed, heavy-duty rotating knives cut unaligned cane stalks into uniform, short billets. This mechanical shear levels out the physical distribution across the primary carrier belt, maintaining a constant mass flow rate.
2. **Secondary Shredding and Cell Disruption:** The uniform billets enter a secondary knife stage linked to a high-inertia hammer shredder. This unit splits the cane stalks longitudinally along their fibrous structures. This process yields fine, highly fiberized cane (*bagacillo*) optimized for extraction, targeting a commercial Preparation Index (PI) greater than 85% open storage cells.
3. **Biochemical Protection at the Feed Gate:** To counter juice loss, the shredded fiber is treated with targeted industrial biocides to halt *Leuconostoc* bacterial strains from fermenting the accessible sucrose. Simultaneously, a continuous dose of dextranase enzymes is applied to digest dextran polymers before the material hits the milling line.

### 4.2.2 Hydraulic Crushing Tandem and Counter-Current Imbibition Architecture

The fiberized raw cane moves sequentially through a heavy 5-mill extraction line. Each individual milling station utilizes three massive, grooved steel rollers (comprising a top roller, feed roller, and discharge roller) executing juice separation under tight hydraulic ram pressures.

- **First-Stage Juice Expressing (Mill 1):** The initial mill processes the incoming fiberized cane without any water addition. It extracts the highest-purity sucrose stream, industrially designated as First Expressed Juice (FEJ), typically recording a dissolved solids content of 18.0° to 21.0° Brix.
- **Reciprocal Imbibition Network:** To recover residual sucrose bound within the dry fiber mat, hot water (maintained strictly between 60°C and 70°C) is sprayed onto the bagasse blanket entering Mill 5.
  - The dilute juice squeezed from Mill 5 (Last Expressed Juice) is redirected upstream to wash the fiber entering Mill 4.
  - This fluid is iteratively pumped backward from Mill 4 to Mill 3, creating an efficient counter-current washing loop. This process strips sucrose

molecules from the moving stalk fiber while keeping fresh water usage to a minimum.

- **Solid Biomass Residue:** The final exhausted fiber mass exiting the fifth mill is discharged as Bagasse. This byproduct exhibits high structural dryness (sucrose content under 2.0% Pol, and roughly 50% retained moisture) and is conveyed immediately to the high-pressure factory boilers for power generation.

#### 4.2.3 Juice Screening, Enzyme Treatment, and Industrial Clarification

The various raw juice fractions recovered from the mill tandem are blended and subjected to chemical and thermal purification to remove suspended field mud, trash, and colloidal proteins.

##### Stream Consolidation and Enzymatic Dosing

- **Primary Feed Blending:** The concentrated First Expressed Juice mixes directly with the recycled dilute streams from the back-end mills to form the main chemical processing fluid.
- **Chemical Treatment:** The moving juice stream is treated with:
  - **Sanitizing Biocides:** To stop sugar-consuming bacterial fermentation.
  - **Amylase Enzymes:** To break down field starch chains that would otherwise increase syrup viscosity down the line.
  - **Dextranase Enzymes:** To neutralize high-molecular-weight polysaccharides.
- **Mechanical De-gritting:** The raw stream runs across fine-mesh rotating screeners (*cush-cush* separators) to trap and remove floating bagasse particles. The strained liquor accumulates in the Mixed Juice surge tanks at a density of 14.0° to 16.0° Brix.

##### Chemical De-acidification and Flocculation

1. **Primary Thermal Elevation:** The acidic mixed juice is pumped through shell-and-tube heat exchangers to bring its baseline temperature up to 70°C.
2. **Alkaline Liming Phase:** Milk of Lime ( $\text{Ca}(\text{OH})_2$ ), modulated to a density of 6.0 to 8.0° Baumé, is metered into the hot juice. This adjusts the natural, acidic juice (pH 5.0–5.4) to a neutral, stable target of pH 7.2–7.6.
  - **Precipitation Reaction:** Reactive calcium ions bind with natural inorganic phosphates inside the juice matrix. This forms an insoluble calcium phosphate precipitate ( $\text{Ca}_3(\text{PO}_4)_2$ ), which acts as a heavy chemical net to trap suspended soil particles and organic colloids:
  - $$3\text{Ca}(\text{OH})_2 + 2\text{H}_3\text{PO}_4 \longrightarrow \text{Ca}_3(\text{PO}_4)_2 + 6\text{H}_2\text{O}$$

3. **Superheating and Flocculant Activation:** The limed liquor is sent to a secondary flash heater under pressure, raising the temperature to 102°C–105°C to degas entrained air bubbles. It is then treated with long-chain anionic polyacrylamide flocculants as it flows into the clarification tanks.
4. **Gravity Separation:** Heavy flocs of trapped mud sink to the base of the continuous clarifiers and are routed to rotary vacuum filters to be discharged as dry Filter Cake waste. The clean, sparkling supernatant liquor overflows the upper weirs as Clear Juice, stabilized at a safe process window of pH 6.8 to 7.2.

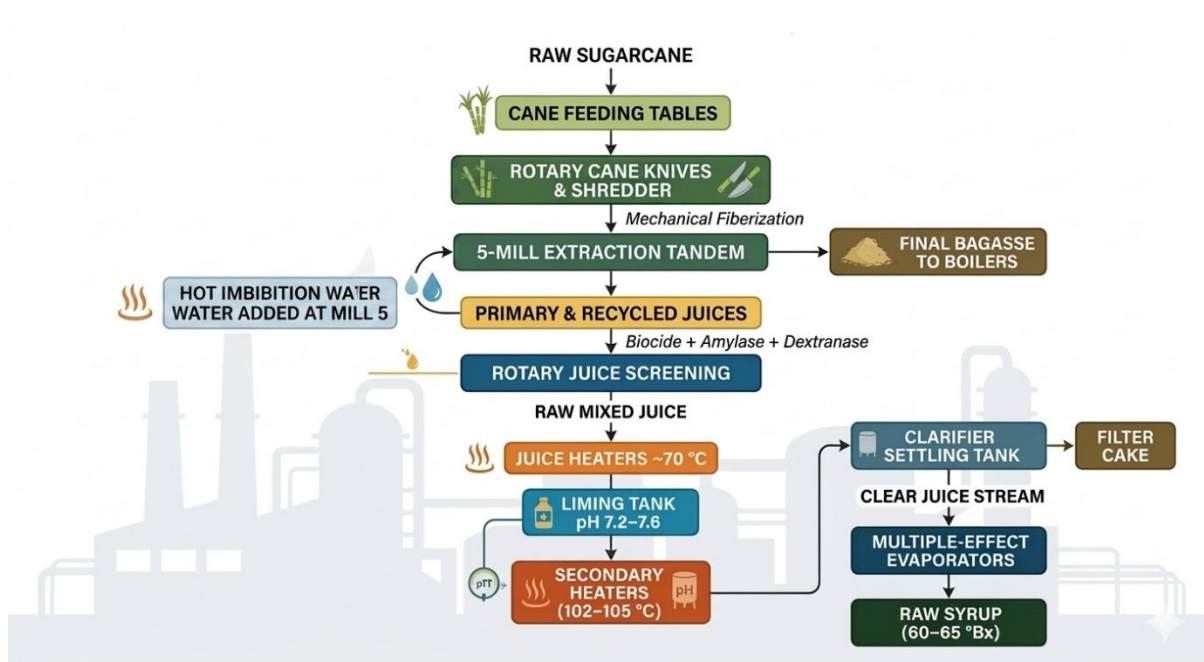


Figure 4: Juice Screening, Enzyme Treatment, and Industrial Clarification

## 4.3 Multiple-Effect Evaporation and Concentrated Syrup Management

### 4.3.1 Principle of Multiple-Effect Evaporation and Thermal Economy

The Clear Juice leaving the clarification sector enters the evaporation station at a relatively dilute concentration of 14.0° to 16.0° Brix. To transform this thin liquid into a crystallizable medium, a massive volume of water must be removed until the dissolved solids density reaches 60.0° to 65.0° Brix.

To accomplish this without excessive energy expenditure, a series of four or five boiling vessels—known as a Multiple-Effect Evaporator (MEE) Tandem—is operated in series.

The primary engineering goal is thermal economy: **one kilogram of fresh steam must evaporate multiple kilograms of water.**

- **Forward Feed Mechanics**

Exhaust steam from the factory's power-generation turbines (typically at 120°C–130°C and a pressure of ~1.5–2.0 bar) is fed exclusively into the heating chamber (*calandria*) of the First Effect.

- **Vapor Re-use**

The water boiled off from the juice in the first vessel becomes the heating medium (vapor) for the second vessel. The vapor from the second vessel heats the third, and so on down the line.

- **The Vacuum Gradient**

Because the boiling point of the juice drops as the pressure drops, a progressive vacuum is drawn across the tandem using barometric condensers and vacuum pumps.

- The First Effect operates under positive pressure at the highest temperature (~115°C–120°C).
- The Final Effect operates under a high vacuum (~0.85 bar) at a much lower temperature (~60°C–65°C), preventing the thermal degradation (caramelization) of sucrose molecules as the syrup becomes highly viscous.

#### **4.3.2 Vapor Bleeding and Energy Integration**

To maximize the overall thermal efficiency of the mill, vapor is not just passed to the next evaporator vessel; it is systematically diverted ("bled") to supply heat to other factory units.

- Vapor Bleeding Pathways:
  - High-pressure vapor from the First Effect (V1) is typically bled to supply heat to the continuous vacuum pans in the crystallization station and the secondary juice flash heaters (105°C).
  - Vapor from the Second Effect (V2) is bled to run the primary juice heaters (70°C).
- This integrated utility system drastically cuts down the total factory exhaust steam demand, leaving more raw bagasse biomass available to be sold or converted into grid electricity.

#### **4.3.3 Operational Degradation: Syrup Inversion and Scale Mitigation**

The evaporation phase poses two major technical challenges that can cause direct product or efficiency losses if left unmanaged:

##### **1. Thermal Inversion Control**

As hot juice sits inside the early high-temperature effects, the combination of heat and any residual acidity can trigger sucrose inversion, splitting valuable sucrose into non-crystallizable glucose and fructose:



- Mitigation: Operators closely monitor residence time and maintain strict chemical control, ensuring the incoming clear juice stays at a slightly alkaline or neutral pH (6.8–7.2). If the pH drops below 6.5, emergency milk of lime dosing is triggered to halt inversion losses.

## 2. Scale Deposition and Antiscalant Dosing

As water is stripped away, less-soluble mineral salts—primarily calcium sulphate ( $\text{CaSO}_4$ ), calcium silicate, and silica—precipitate out of the solution. These minerals bake onto the internal walls of the stainless steel evaporator tubes, forming a hard, insulating crust known as scale.

- The Problem: Scale rapidly destroys heat transfer efficiency. If unmanaged, steam consumption spikes, and the factory throughput drops because the juice cannot boil fast enough.
- Mitigation: Continuous chemical dosing pumps inject specialized polyacrylate or organophosphonate antiscalants into the juice stream entering the 3rd and 4th effects. These polymers distort the crystal lattice of the precipitating minerals, preventing them from baking onto the tube walls and extending the time between mechanical shutdowns for acid-washing.

### 4.3.4 Syrup Clarification and Storage Sulfitation

The highly concentrated product leaving the final evaporator effect is a thick, dark golden liquid called Raw Thick Syrup (60.0°–65.0° Brix). Before it can be routed to the crystallization pans, it must undergo a final purification step.

#### 1. Syrup Sulfitation (Bleaching)

The thick syrup is sent through a sulfitation tower where it is exposed to sulfur dioxide ( $\text{SO}_2$ ) gas generated by burning pure sulfur blocks.

- Chemical Action: The dissolved gas acts as a reducing agent, bleaching out natural organic colorants (polyphenols and melanoidins) to ensure the final white sugar crystals meet international color index standards (ICUMSA specifications). It also slightly lowers the viscosity of the syrup for easier boiling.

#### 2. Surge Storage

The clarified, bleached thick syrup is pumped directly into insulated storage tanks equipped with continuous brix controllers, ready to be drawn into the vacuum pans for the crystallization stage.

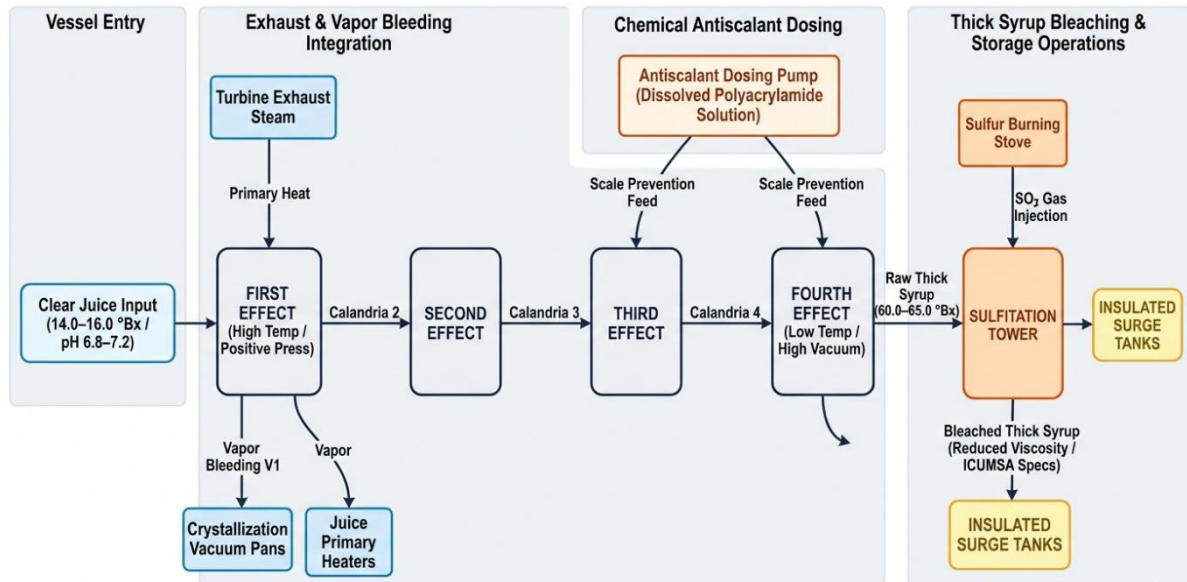


Figure 5: Syrup Clarification and Storage Sulfitation

## 4.4 Vacuum Pan Crystallization, Strike Receivers, and Centrifugal Separation

### 4.4.1 Mechanics of Vacuum Pan Boiling and Fractional Molasses Separation

The transformation of sulfited thick syrup into high-purity sugar crystals relies on a controlled, step-by-step boiling and curing process. Because high temperatures cause sucrose to decompose, single-effect **Vacuum Pans** are run under a tight vacuum ( $\sim 0.85$  bar), dropping the boiling temperature to  $65^{\circ}\text{C}$ – $70^{\circ}\text{C}$ .

To extract the maximum amount of sugar and minimize losses, factories use a strict **Three-Boiling (A, B, and C) Sugar Crystallization Scheme**. This scheme separates the run-off from the centrifuges into distinct **Heavy** and **Light** fractions based on purity

- **A-Boiling & Fractional Run-offs**

The A-Vacuum Pan is fed a mixture of fresh thick syrup, high-grade melted sugar fractions, and B-Light molasses. It is cooked into a dense crystal slurry called **A-Massecuite**.



- When spun in the centrifugals, the initial high-purity liquor thrown off before washing is collected as **A-Heavy Molasses** (routed directly to feed the B-Vacuum Pan).
  - The subsequent wash-water run-off is captured separately as **A-Light Molasses** (recycled back to feed the A-Vacuum Pan to save high-purity sucrose).
- **B-Boiling & Fractional Run-offs**

The B-Vacuum Pan uses a crystalline base ("footing") made from recycled C-sugar, fed primarily by **A-Heavy Molasses**. It is boiled into **B-Massecuite**.

    - The initial spin without water yields **B-Heavy Molasses** (routed downstream to feed the C-Vacuum Pan).
    - Washing the intermediate crystals produces **B-Light Molasses** (recycled back upward to the A-Vacuum Pan feed lines).
  - **C-Boiling & Final Exhaustion**

The lowest-purity boiling stage takes place in the C-Vacuum Pan, which is fed by **B-Heavy Molasses** and concentrated into **C-Massecuite**.

    - The initial spinning of this low-grade material yields completely exhausted **Final Molasses**, which is discharged out of the factory process loop as a waste byproduct.
    - Any minor washing steps or curing run-offs yield **C-Light Molasses**, which is kept inside the loop to melt or condition low-grade recovery sugars.

#### 4.4.2 True Slurry Dosing: Micro-Crystal Pan Seeding

Uncontrolled boiling causes sugar crystals to grow spontaneously in erratic sizes, ruining the uniform layout required for commercial packaging. To enforce a uniform grain size, operators utilize a technique called **True Slurry Seeding**:

- **The Seed Slurry Composition**

A highly controlled slurry consisting of micro-fine sugar crystal nodes ground to a size of 1–5 microns, suspended in pure, anhydrous isopropyl or ethyl alcohol.
- **The Injection Process:** When the boiling medium inside any vacuum pan reaches a specific supersaturation point, a calibrated volume of this seed slurry is injected directly through a pneumatic dosing valve.
- **The Physical Action:** The alcohol vehicle flashes off instantly into the vacuum atmosphere, dispersing the millions of microscopic crystal nodes evenly

throughout the boiling volume. Each tiny node acts as a physical seed node onto which dissolved sucrose molecules precipitate directly, ensuring all crystals grow uniformly to the targeted grain diameter.

#### 4.4.3 Strike Receivers and Crystal Conditioning

Once a vacuum pan cycle reaches its volumetric limit and optimal crystal-to-liquor ratio, the bottom discharge valve is opened. The pan contents are dropped ("struck") into massive holding vessels called **Strike Receivers** or **Crystallizers**.

- **Mechanical Agitation:** Strike receivers are equipped with slow-moving mechanical stirrers to keep the heavy massecuite mass from solidifying into a concrete-like block.
- **Thermal Conditioning (Cooling Crystallization):** C-massecuites, in particular, are kept in these receivers for up to 24–48 hours to undergo progressive cooling. As the temperature drops slowly from 70°C down to 40°C, the solubility of sucrose decreases further. This forces remaining dissolved sugar molecules out of the liquid matrix and onto the existing crystal walls, completing the final exhaustion of the mother liquor before separation.

#### 4.4.4 Centrifugal Separation and High-Gravity Curing

The conditioned, thick massecuite slurry is pumped directly to a battery of basket **Centrifugals** to mechanically isolate the solid sugar crystals from the surrounding liquid molasses fractions using high centrifugal force.

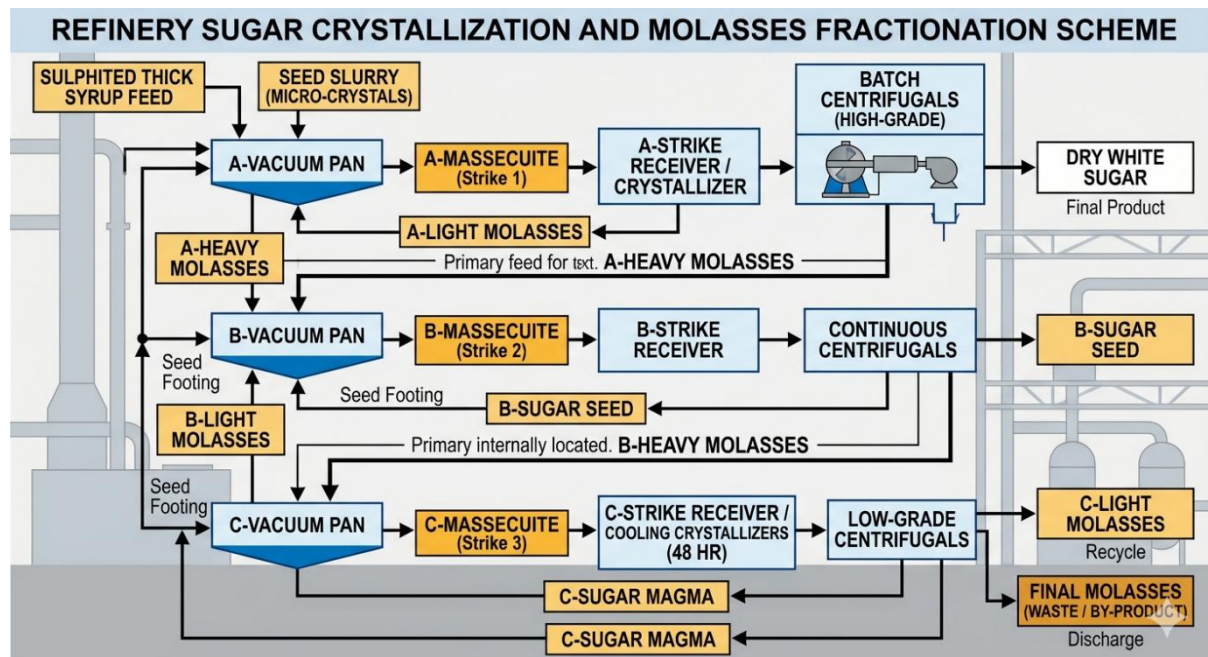


Figure 6: Centrifugal Separation and High-Gravity Curing

- **Batch Centrifugals (High-Grade A-Sugar)**

A-massecuite is fed into a perforated cylindrical stainless steel basket that spins at high speeds (1,200 to 1,500 RPM). Extreme centrifugal force throws the liquid molasses outwards through the fine mesh walls, while the solid sugar crystals remain trapped inside. A brief spray of superheated water washes the crystal cake to remove residual surface molasses before a mechanical scraper plows the dry white sugar out the bottom.

- **Continuous Centrifugals (Low-Grade B & C Sugars)**

Low-grade massecuities run through conical-shaped continuous centrifugals. The material flows up the spinning cone walls; molasses passes through the screen, and the separated crystals continuously spill over the upper edge for processing or recycling.

## **4.5 Sugar Drying, Conditioning, and Packaging Operations**

### **4.5.1 Mechanics of Rotary Cascade Dryers and Moisture Equilibrium**

The high-purity white sugar crystals discharged from the batch centrifugals are physically isolated but still retain a surface layer of moisture (approximately 0.5% to 1.5% water by weight). If this wet sugar is bagged immediately, it will cake into solid blocks due to moisture migration.

To stabilize the product, the wet crystals are fed continuously into a **Rotary Cascade Dryer and Cooler Tandem** (commonly called a sugar drum)

- **The Cascading Action**

The dryer consists of a massive, slightly inclined rotating steel cylinder fitted with internal lifters (*flights*). As the drum rotates, these flights scoop up the wet sugar and drop it in a continuous, uniform curtain (cascade) through the length of the cylinder.

- **Dual-Stage Air Interaction**

- **The Drying Zone:** In the first half of the drum, a counter-current stream of hot, filtered air (65°C to 75°C) passes through the falling sugar curtain. This evaporates the surface moisture down to a tight threshold of **<0.04%**.
- **The Cooling Zone:** In the second half, co-current ambient air (30°C) cools the hot crystals down to within 5°C of room temperature. If sugar

is packed hot, it will absorb ambient moisture later inside the bag as it cools, leading to crystallization binding (clumping).

#### 4.5.2 Quality Metrics: ICUMSA Color Scores and Safety Limits

The final sugar product must meet strict national and international regulatory benchmarks before warehouse dispatch. The primary technical parameter used to grade the efficiency of the upstream clarification and sulfitation lines is the **ICUMSA Color Score** (International Commission for Uniform Methods of Sugar Analysis):

$$\text{ICUMSA Value (IU)} = A_{420} / (b \times c) \times 10^5$$

Where:

- $A_{420}$  = Optical absorbance of the sugar solution at a light wavelength of 420 nm.
- $b$  = Cell path length (cm).
- $c$  = Solution concentration (g/mL).

#### Commercial Grade Boundaries & Quality Tolerances:

1. **Premium White Sugar (Grade 1):** Target **< 45 IU**. This represents pure white refined sugar with all organic colorants completely bleached out or removed via clarification.
2. **Standard Mill White Sugar (Grade 2):** Target **45 – 150 IU**. This is the standard commercial table sugar produced at Nzoia Sugar Company, showing a clean white appearance with minute trace fractions of natural molasses.
3. **Brown / Raw Sugar (VHP):** Target **> 400 IU**. High-color sugar skipped from the sulfitation and fine washing loops, retaining a thick molasses coating.
4. **Safety Parameters:** Total invert sugar content must remain under **0.04%**, and ash content must not exceed **0.02%** to pass export shelf-life certification.

#### 4.5.3 Grading Screens, Magnetic Separation, and Automated Bagging

Before hitting the storage hoppers, the dried crystals pass through mechanical sizing and safety screening networks to remove anomalies:

1. **Mechanical Vibratory Graders:** The sugar flows over a series of multi-deck vibrating wire screens. This separates the output into three physical fractions: **Oversized lumps** (sent back to be dissolved and re-boiled), **Product-grade crystals** (0.6mm to 1.0mm diameter), and **Fine dust** (recycled into the pan house footing).
2. **Rare-Earth Magnetic Traps:** The graded sugar falls through high-intensity rare-earth magnets positioned inside the chutes. This traps any trace microscopic metal shavings or iron scale picked up from the moving conveyor belts or dryer walls.

3. **Automated Bagging & Stitching:** Graded sugar enters the packaging lines, feeding automated load-cell scales. The sugar is dropped into 50kg or 1kg woven polypropylene bags lined with polyethylene moisture barriers. The bags are automatically stitched closed and moved via conveyors directly to the holding warehouse.

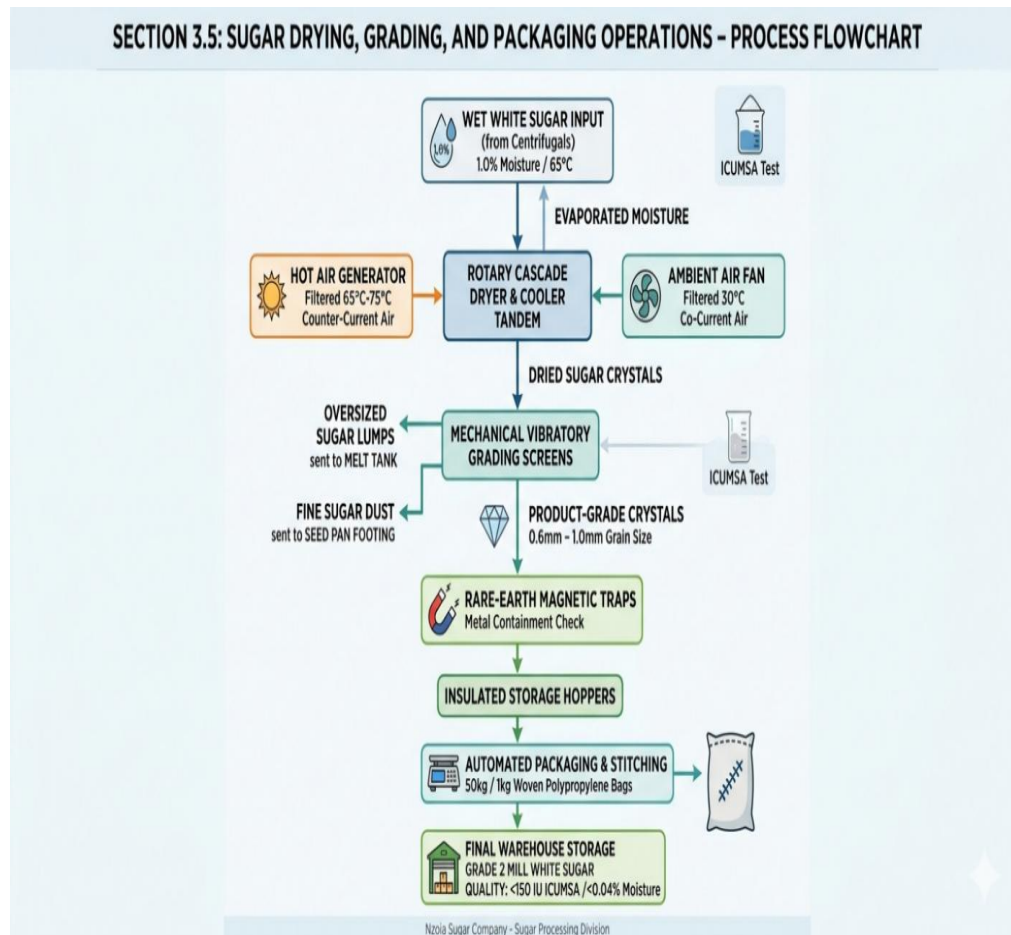


Figure 7: Grading Screens, Magnetic Separation, and Automated Bagging

## 4.6 Byproduct Utilization and Co-Generation Framework

### 4.6.1 Industrial Mass Balance and Byproduct Yields

The milling and processing of sugarcane generates three distinct solid and liquid structural byproducts. Rather than traditional waste discharge, modern sugar mills operate as integrated biorefineries. Every metric ton of cane crushed yields a highly predictable ratio of valuable materials:

- **Bagasse (Fibrous Biomass):** Yields **28% to 30%** of total crushed cane mass.
- **Filter Cake (Filter Mud):** Yields **3.5% to 4.5%** of total crushed cane mass.
- **Final Molasses (Exhausted Liquor):** Yields **3.0% to 3.5%** of total crushed cane mass.

#### 4.6.2 Bagasse-Fired Co-Generation and Thermal Energy Systems

- The fibrous waste material discarded by the final mill stage is transferred directly via elevated conveyors to the factory's power facility.
1. **High-Pressure Combustion:** The wet bagasse (containing ~50% moisture) is fed directly into high-pressure water-tube boilers. It burns at high temperatures to generate superheated steam at a pressure of **45 to 65 bar** and temperatures near **450°C**.
  2. **Turbo-Alternator Electricity Production:** This high-pressure steam expands through multi-stage extraction-condensing steam turbines linked to electrical alternators. The system generates all the electrical power required to run the mill's heavy motors, shredders, and centrifugals.
  3. **Low-Pressure Steam Recycling:** The low-pressure exhaust steam leaving the turbine stage (~1.5 to 2.0 bar) is not wasted. It is piped back to the processing plant to supply heat to the multiple-effect evaporators and vacuum boiling pans.

#### 4.6.3 Filter Cake Management: Nutrient Recycling

The heavy, solid mud settled out by continuous clarifiers and scraped off the rotary vacuum filters is designated as Filter Cake.

- **Nutrient Rich Profile:** This material contains concentrated amounts of nitrogen, phosphorus, potassium, organic carbon, and trace minerals scraped off the incoming cane stalks.
- **Agricultural Redistribution:** The factory stores this mud in open-air collection bays. It is distributed back to contracted outgrower farmers to be used as an organic soil conditioner. This reduces the farmers' reliance on expensive commercial chemical fertilizers and improves the soil structure in the outgrower farming zone.

#### 4.6.4 Final Molasses Distribution: Downstream Industrial Uses

The dark, highly viscous liquid discharged from the continuous C-centrifugals contains zero crystallizable sucrose but is rich in reducing sugars (glucose and fructose).

- **Industrial Distilleries:** The molasses is pumped into large steel storage tanks and sold to external chemical distilleries. It is used as the primary raw material for industrial fermentation to produce fuel-grade ethanol, extra neutral alcohol (ENA), and industrial spirits.
- **Animal Feed Formulations:** Due to its high carbohydrate density and palatability, large fractions are purchased by agricultural feed mills. They mix it

with dry crop residues to manufacture enriched liquid cattle feeds and stock blocks.

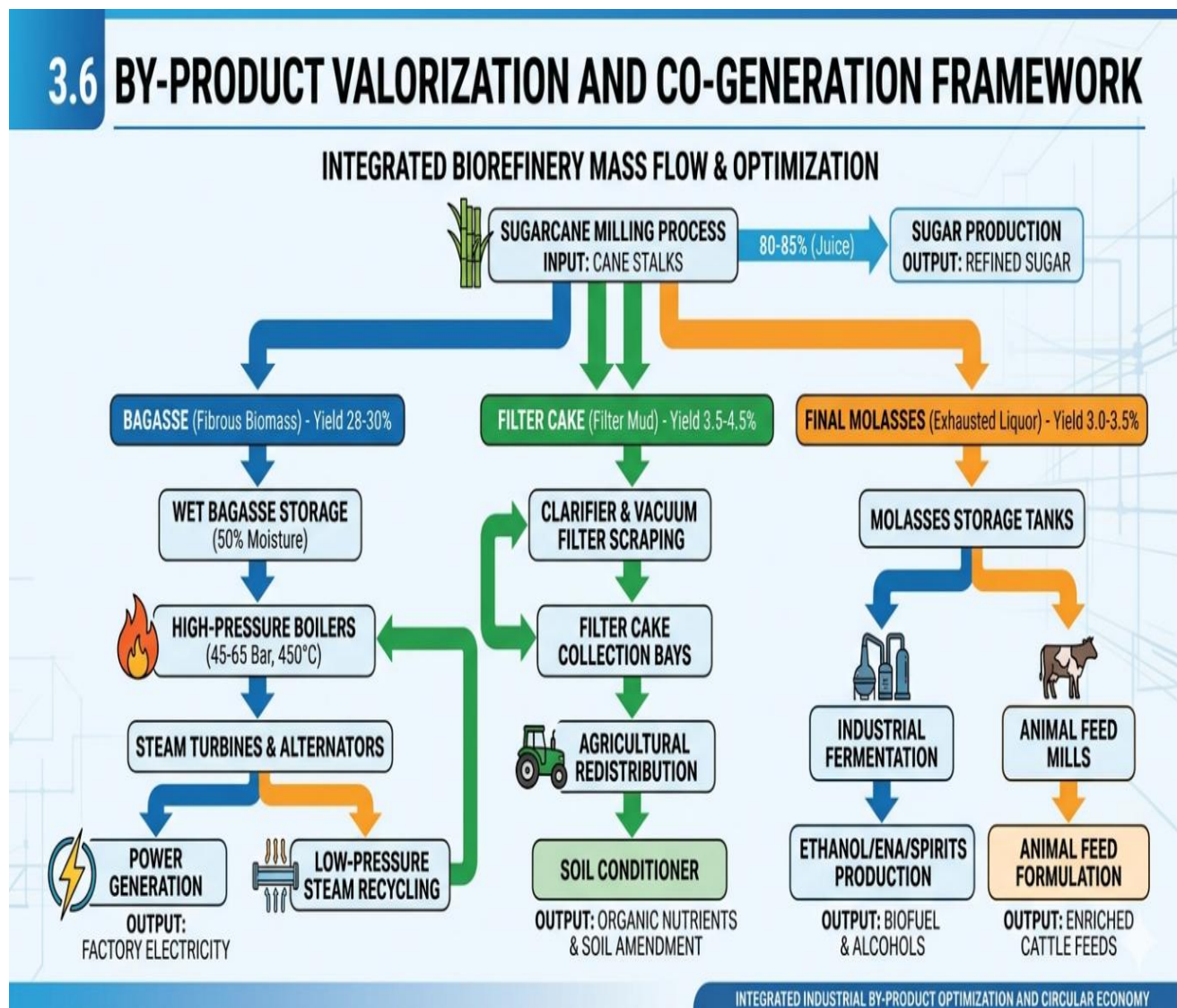


Figure 8: Final Molasses Distribution: Downstream Industrial Uses



# CHAPTER 5: NZOIA SUGAR COMPANY UTILITY WATER TREATMENT & WASTEWATER MANAGEMENT.

## 5.1 WATER INTAKE, RAW STORAGE, AND PRE-SEDIMENTATION

Nzoia Sugar Company relies on raw water drawn continuously from the **River Kuywa**. Raw surface water contains varying levels of suspended silt, clay, organic detritus, and microbiological contaminants, requiring physical and chemical treatment prior to industrial or domestic utility distribution.

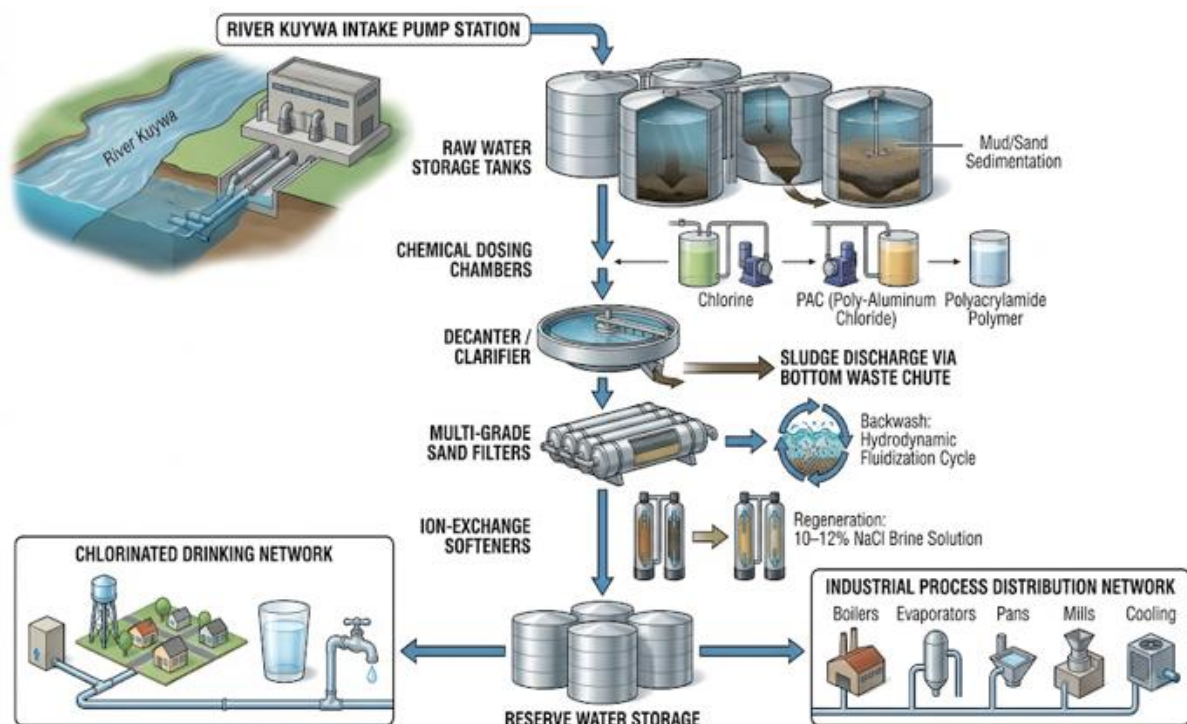


Figure 9: water intake, raw storage, and pre-sedimentation

1. **Intake Pumping:** Raw water is abstracted from River Kuywa via submersed intake pumps protected by mechanical trash racks to block heavy debris, branches, and large stones.
2. **Primary Sedimentation in Raw Water Tanks:** Water enters large-capacity raw water storage tanks designed to serve a dual function: bulk liquid storage and primary gravity settling. The reduced linear velocity within the tanks allows dense suspended solids (heavy sand, gravel, and silt) to settle to the bottom floor, reducing the turbidity load on downstream equipment.



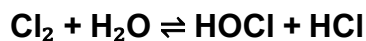
## 5.2 CHEMICAL PRE-TREATMENT, COAGULATION, AND DECANter CLARIFICATION

### 5.2.1 CHEMICAL DOSING & PRE-OXIDATION

Water exiting the primary settling tanks is treated with chemical reagents to destabilize colloidal suspensions and eliminate organic matter:

- **Chlorine (Cl<sub>2</sub>) / Sodium Hypochlorite (NaOCl):** Primary disinfectant and oxidizing agent. Pre-chlorination oxidizes dissolved iron and manganese, bleaches natural organic color compounds, prevents bio-fouling on filter media, and adjusts the raw water pH into an optimal working window of **pH 6.8–7.2**.

**Chemical Reaction:**



- **Poly-Aluminum Chloride (PAC, [Al<sub>2</sub>(OH)<sub>n</sub>Cl<sub>6-n</sub>]<sub>m</sub>):** An inorganic primary coagulant. Colloidal clay particles carry negative surface charges that prevent them from coalescing. High-charge **Al<sup>3+</sup>** polymeric cations neutralize these negative charges, allowing micro-flocs to form.
- **Polyacrylamide (PAM) Polymer Catalyst:** High-molecular-weight anionic polyacrylamide polymer acts as a flocculant aid. Long polymer chains bind micro-flocs together into heavy, dense aggregates through inter-particle bridging.

### 5.2.2 DECANter CLARIFICATION AND SLUDGE REMOVAL

The chemically treated water flows directly into the **Decanter (Clarifier Unit)**, where four concurrent physical-chemical operations occur:

1. **Coagulation:** Rapid mixing neutralizes particle surface charges.
2. **Flocculation:** Slow mixing promotes collision and growth of polymer-bound flocs.
3. **Sedimentation:** Dense flocs settle out of the upward-flowing water column toward the conical floor.
4. **Clarification:** Purified water overflows into perimeter launders while settled sludge accumulates in a bottom hopper and is discharged through a **waste sludge chute**.

## 5.3 MULTI-GRADE SAND FILTRATION (MGSF) AND BACKWASHING DYNAMICS

### 5.3.1 BED MEDIA STRATIFICATION AND GRADED MEDIA SETUP

Water from the decanter passes through **Multi-Grade Sand Filters (MGSF)** to remove residual carry-over flocs and fine suspended solids down to < 5 microns.

Unlike conventional single-media sand filters that clog rapidly at the top surface layer, an MGSF utilizes coarse-to-fine stratification across depth, utilizing the full volume of the bed.

### Multi-Grade Sand Filter Media Bed Layering Specifications

| Bed Stratum                    | Media Composition        | Layer Depth (Inches / mm) | Particle Size Range | Primary Functional Role                                    |
|--------------------------------|--------------------------|---------------------------|---------------------|------------------------------------------------------------|
| <b>Top Layer (Filter Zone)</b> | Fine Silica Sand         | 12 in (300 mm)            | 0.6–1.2 mm          | Final straining of fine suspended particles                |
| <b>Mid Layer (Pre-Filter)</b>  | Coarse Sand / Anthracite | 10 in (250 mm)            | 1.2–2.4 mm          | Bulk deep-bed detention of larger flocs                    |
| <b>Sub-Base Layer</b>          | Fine Gravel / Pebbles    | 6 in (150 mm)             | 3.0–6.0 mm          | Transition zone preventing sand ingress to underdrain      |
| <b>Base Support Layer 1</b>    | Medium Gravel            | 6 in (150 mm)             | 12.0–18.0 mm        | Mechanical weight distribution layer                       |
| <b>Bottom Base Layer 2</b>     | Coarse Gravel            | 8 in (200 mm)             | 25.0–50.0 mm        | Encases bottom lateral strainers; uniform water collection |

Table 1: Multi-Grade Sand Filter Media Bed Layering Specifications

### 5.3.2 BACKWASH MECHANISM AND HYDRODYNAMIC FLUIDIZATION

Over time, entrapped solids build up in the sand bed, causing channel formation and increasing differential pressure ( $\Delta P > 0.8$  bar).

To restore flow capacity, the filter undergoes a **Backwashing Cycle**:

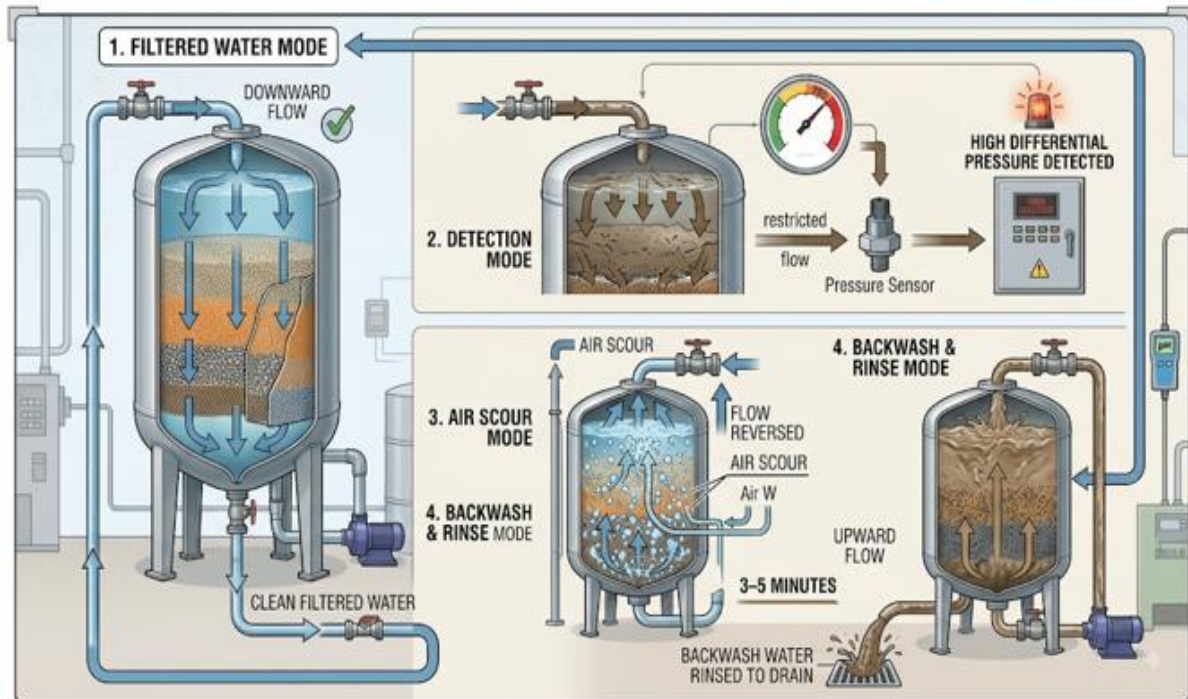


Figure 10: Backwash mechanism and hydrodynamic Fluidization

1. **Isolation:** The raw water inlet and filtered water outlet valves are closed.
2. **Air Scouring:** High-pressure air (1.5–2.0 bar) is blown upward through the bottom underdrain laterals for 3–5 minutes. The expanding air bubbles rub the sand grains against each other, scouring off attached mud clumps and organic films.
3. **Water Backwashing (Bed Fluidization):** Filtered water is pumped in reverse (bottom-to-top) at high velocity (30–40 m<sup>3</sup>/m<sup>2</sup>/hr). This upward flow expands the sand bed by 30–50%, fluidizing the media and floating off liberated dirt into the top backwash discharge trough.
4. **Rinse/Settling:** Flow direction is returned to normal (top-to-bottom) for 5 minutes to settle media layers back into their stratified positions before returning the unit online.

## 5.4 ION-EXCHANGE SOFTENING, RESIN CHEMISTRY, AND BRINE REGENERATION

### 5.4.1 SOFTENER UNIT ARCHITECTURE

Filtered Water exiting the MGSFs contains dissolved hardness minerals—principally calcium (Ca<sup>2+</sup>) and magnesium (Mg<sup>2+</sup>) ions. If unmanaged, hard water forms insulating scale inside

steam boiler tubes, leading to tube failures and heat transfer loss. Nzoia Sugar operates 3 cylindrical softener vessels. Each vessel contains a base layer of fine gravel supporting a thick bed of Strong Acid Cation (SAC) exchange polymer resin.

### 5.4.2 POLYMER RESIN CHEMICAL STRUCTURE AND EXCHANGE MECHANISM

The softening resin consists of spherical beads constructed from a Polystyrene-Divinylbenzene (PS-DVB) copolymer matrix, functionalized with negatively charged sulfonic acid groups ( $-\text{SO}_3^-$ ) pre-loaded with exchangeable sodium ions ( $\text{Na}^+$ ).

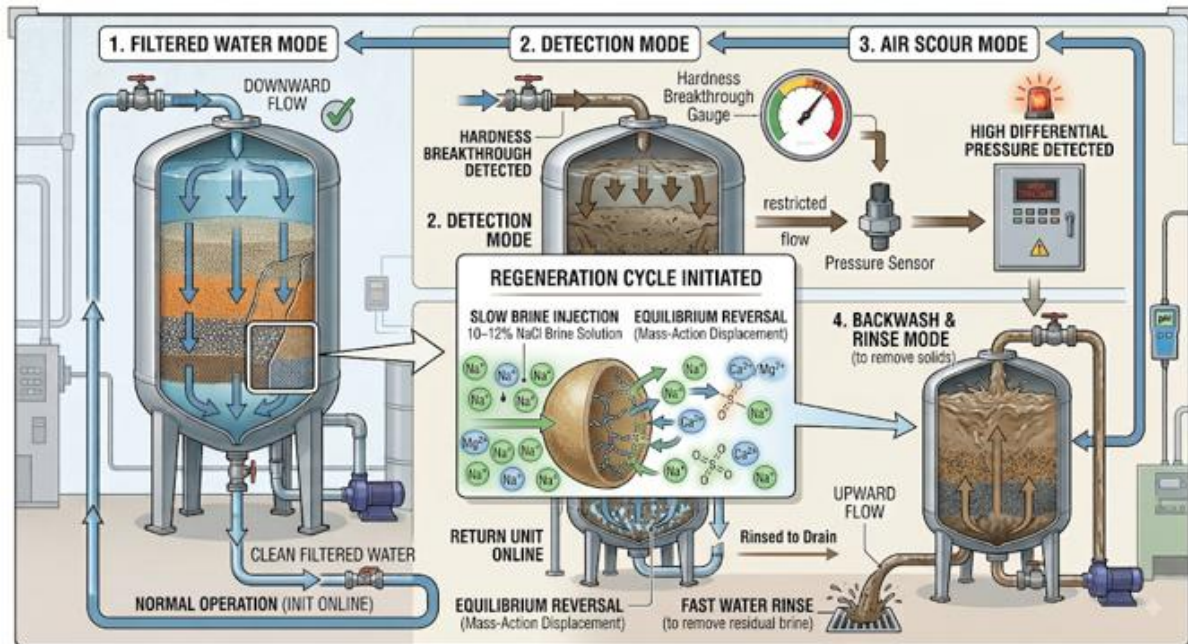
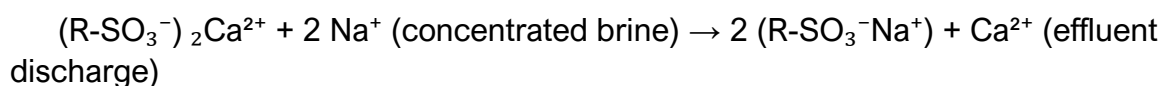


Figure 11: Polymer Resin Chemical Structure and Exchange Mechanism

- **Backwash Phase:** Water is passed upward to loosen packed resin beads and remove dirt.
- **Brine Injection (Mass-Action Reversal):** A saturated **10–12% NaCl industrial salt solution** is slowly drawn into the vessel. The excess concentration of  $\text{Na}^+$  ions overrides the resin's thermodynamic preference for divalent ions, driving the exchange reaction in reverse:



- **Slow Rinse:** Distilled water flushes remaining displaced  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ , and chloride ions to the sewer.
- **Fast Rinse:** High-rate rinse flushes trace salt out until discharge conductivity drops to normal levels, restoring the unit to service.

## 5.5 PLANT WATER DISTRIBUTION AND INTERNAL POLISHING NETWORK

Softened and filtered waters are stored in designated tanks and distributed across factory operations:

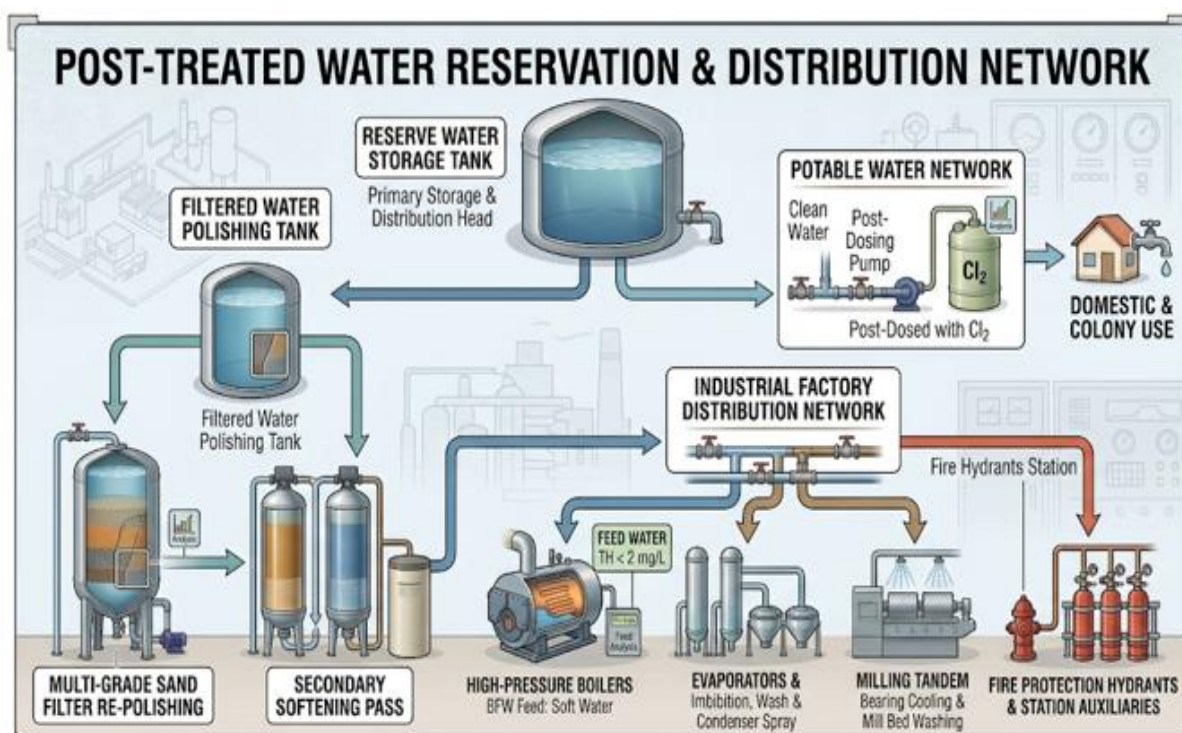


Figure 12: Plant water distribution and internal polishing network

- **Potable / Drinking Water Loop:** Water from the reserve storage tank is directed through a secondary chlorination step to maintain a residual chlorine concentration of 0.5–1.0 mg/L before distribution to staff quarters, offices, and lab drinking fountains.
- **Factory Process Loop:** Filtered water supplies mill journal bearing cooling, rotary spray nozzles, and juice dilution stations.
- **High-Purity Boiler Feed Loop:** Softened water undergoes secondary polishing to remove dissolved gases (via thermal deaeration at 105°C before being pumped to high-pressure steam boilers.

## 5.6 WATER QUALITY SPECIFICATIONS AND SHIFT DATA TABLE

The table below outlines target parameters across water treatment stages alongside representative analytical bench data recorded during shift operations.

Master Shift Analytical Data Table for Water Treatment Stations

| Water Quality Parameter and Test Method | Raw Water (River Kuywa) | Clarified Water (Decanter) | MGSF Exit (Filtered) | Softener Exit (Softened) | Boiler Feed Water (Polished BFW) |
|-----------------------------------------|-------------------------|----------------------------|----------------------|--------------------------|----------------------------------|
|                                         |                         |                            |                      |                          |                                  |

|                                                |                                |                                   |                             |                                 |                                         |
|------------------------------------------------|--------------------------------|-----------------------------------|-----------------------------|---------------------------------|-----------------------------------------|
| <b>pH (Glass Electrode)</b>                    | 6.45                           | 7.10                              | 7.05                        | 7.20                            | 8.80–9.20*                              |
| <b>Turbidity (NTU)</b>                         | 145.0 NTU                      | 8.50 NTU                          | 0.85 NTU                    | 0.40 NTU                        | <0.10 NTU                               |
| <b>Total Dissolved Solids (TDS)</b>            | 180.0 mg/L                     | 210.0 mg/L                        | 205.0 mg/L                  | 215.0 mg/L                      | <30.0 mg/L                              |
| <b>Total Hardness (as CaCO<sub>3</sub>)</b>    | 112.0 mg/L                     | 108.0 mg/L                        | 106.0 mg/L                  | 1.80 mg/L                       | 0.00 mg/L                               |
| <b>Water Quality Parameter and Test Method</b> | <b>Raw Water (River Kuywa)</b> | <b>Clarified Water (Decanter)</b> | <b>MGSF Exit (Filtered)</b> | <b>Softener Exit (Softened)</b> | <b>Boiler Feed Water (Polished BFW)</b> |
| <b>Calcium Hardness (Ca<sup>2+</sup>)</b>      | 78.0 mg/L                      | 75.0 mg/L                         | 74.0 mg/L                   | 1.20 mg/L                       | 0.00 mg/L                               |
| <b>Magnesium Hardness (Mg<sup>2+</sup>)</b>    | 34.0 mg/L                      | 33.0 mg/L                         | 32.0 mg/L                   | 0.60 mg/L                       | 0.00 mg/L                               |
| <b>Free Residual Chlorine (Cl<sub>2</sub>)</b> | 0.00 mg/L                      | 0.80 mg/L                         | 0.45 mg/L                   | 0.30 mg/L                       | 0.00 mg/L                               |
| <b>Chemical Dosing Applied</b>                 | None                           | PAC + Polymer                     | None                        | NaCl Brine (Regen)              | Trisodium Phosphate                     |

**Note on Boiler Feed Water pH: BFW pH is adjusted to 8.80–9.20 using trisodium phosphate (Na<sub>3</sub>PO<sub>4</sub>) dosing to protect boiler steel tubes from acidic corrosion**

Table 2: Master Shift Analytical Data Table for Water Treatment Stations



# CHAPTER 6: EFFLUENT TREATMENT PLANT (ETP) AND ENVIRONMENTAL CONTROL

## 6.1 OVERVIEW AND MASTER FLOW SCHEME OF FACTORY EFFLUENT TREATMENT

Sugar manufacturing generates substantial volumes of liquid effluent containing residual sugars, floor washings, boiler blowdown, mill lubricants, and inorganic ash. Raw effluent possesses a high biological and chemical oxygen demand alongside acidic pH characteristics. Discharging untreated waste directly into local water bodies would deplete dissolved oxygen, causing aquatic die-offs and environmental degradation.

Nzoia Sugar Company operates a continuous multi-stage **Effluent Treatment Plant (ETP)** that combines physical separation, chemical conditioning, anaerobic digestion, and biological phytoremediation to purify factory wastewater before discharging it back into the **River Kuywa**

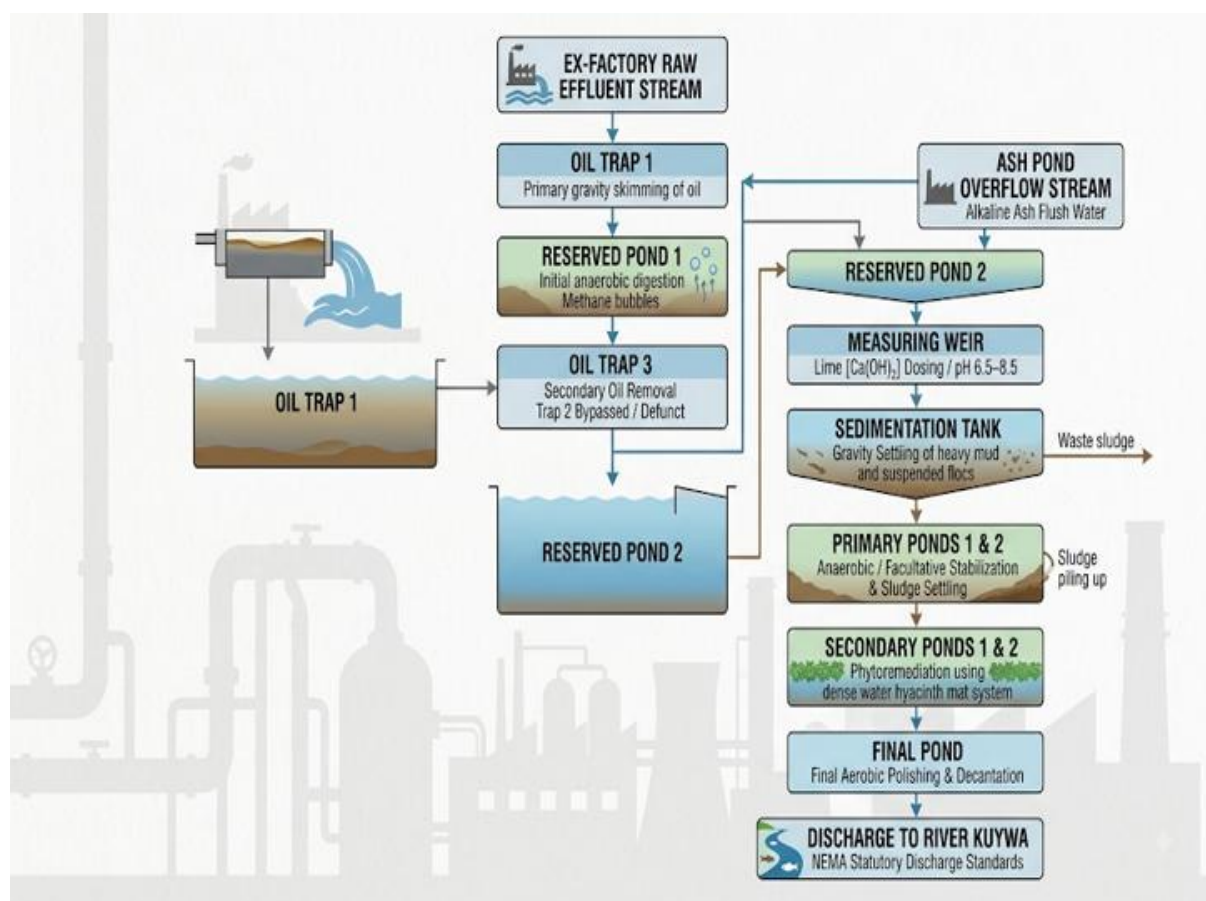


Figure 13: Master flow scheme of factory effluent treatment

## 6.2 PRIMARY PHYSICAL TREATMENT AND OIL SEPARATION DYNAMICS

### 5.2.1 OIL TRAP OPERATIONS AND DENSITY SEPARATION MECHANICS

Raw effluent leaving the factory floor carries lubricating oils, greases, and hydraulic fluids washed down from the mill turbine bearings and heavy machinery. Oil forms an impermeable surface film over water ponds, blocking atmospheric oxygen transfer and suffocating aerobic microorganisms.

Oil traps operate on **gravity-driven phase separation** governed by **Stokes' Law**, utilizing differential mass densities between immiscible liquids:

Density of Water ( $\rho_w \approx 1000 \text{ kg/m}^3$ ) > Density of Industrial Oil ( $\rho_o \approx 850\text{--}900 \text{ kg/m}^3$ )

- **Mechanism:** Wastewater enters the baffle chambers of **Oil Trap 1** at a low horizontal flow velocity. The lower density of oil causes floating globules to coalesce and rise to the liquid surface. Baffle walls extending below the surface trap the floating oil layer while allowing clarified water to pass underneath the baffle lip.
- **Secondary Trapping:** Effluent flows from Reserved Pond 1 into **Oil Trap 3** (Oil Trap 2 is permanently offline due to mechanical failure) to catch residual traces of emulsified oils prior to downstream biological treatment. Skimmed oil is removed manually into waste drums.

### 6.2.2 RESERVED PONDS AND ANAEROBIC DIGESTION DYNAMICS

Effluent exiting Oil Trap 1 passes into **Reserved Pond 1** and subsequently **Reserved Pond 2**. These deep, un-aerated earthen basins function as primary anaerobic stabilization zones.

- **Biochemical Process:** Facultative and obligate anaerobic bacteria break down high-concentration dissolved sucrose and complex organic carbohydrates in the absence of dissolved oxygen ( $\text{DO} = 0.0 \text{ mg/L}$ ).
- **Acidogenesis and Methanogenesis:** Organisms convert complex sugars into Volatile Fatty Acids (VFAs, e.g., acetic, propionic, and butyric acids) and subsequently into methane ( $\text{CH}_4$ ), carbon dioxide ( $\text{CO}_2$ ), and hydrogen sulfide ( $\text{H}_2\text{S}$ ).
- **Odor Generation:** Anaerobic degradation generates a characteristic pungent odor (rotten-egg smell) due to trace  $\text{H}_2\text{S}$  emission and acid generation, confirming active anaerobic digestion and organic load reduction.

### 6.2.3 ASH POND RECIRCULATION AND INTEGRATION

Water utilized in the boiler house to sluice out bottom ash and fly ash from the boiler furnaces is routed to a dedicated Ash Pond.

- **Settling and Alkaline Contribution:** The ash pond acts as a primary gravity settling basin for heavy inorganic furnace ash particles. Uncombusted biomass ash contains alkaline earth oxides and carbonates ( $\text{K}_2\text{O}$ ,  $\text{CaO}$ ,  $\text{Na}_2\text{O}$ ).
- **pH Elevation:** The overflow liquid exiting the ash pond is strongly alkaline ( $\text{pH } 8.5 - 10.0$ ). Routing this alkaline stream into the main effluent line prior to the measuring



weir neutralizes the natural acidity of raw factory effluent (pH 4.5 – 5.5), reducing downstream chemical lime consumption.

## 6.3 CHEMICAL CONDITIONING, SEDIMENTATION, AND PRIMARY STABILIZATION

### 6.3.1 MEASURING WEIR AND LIME DOSING OPERATIONS

The combined effluent stream flows through a **Parshall Flume / V-Notch Measuring Weir** that serves as both a hydraulic flow rate measurement point and a chemical mixing station.

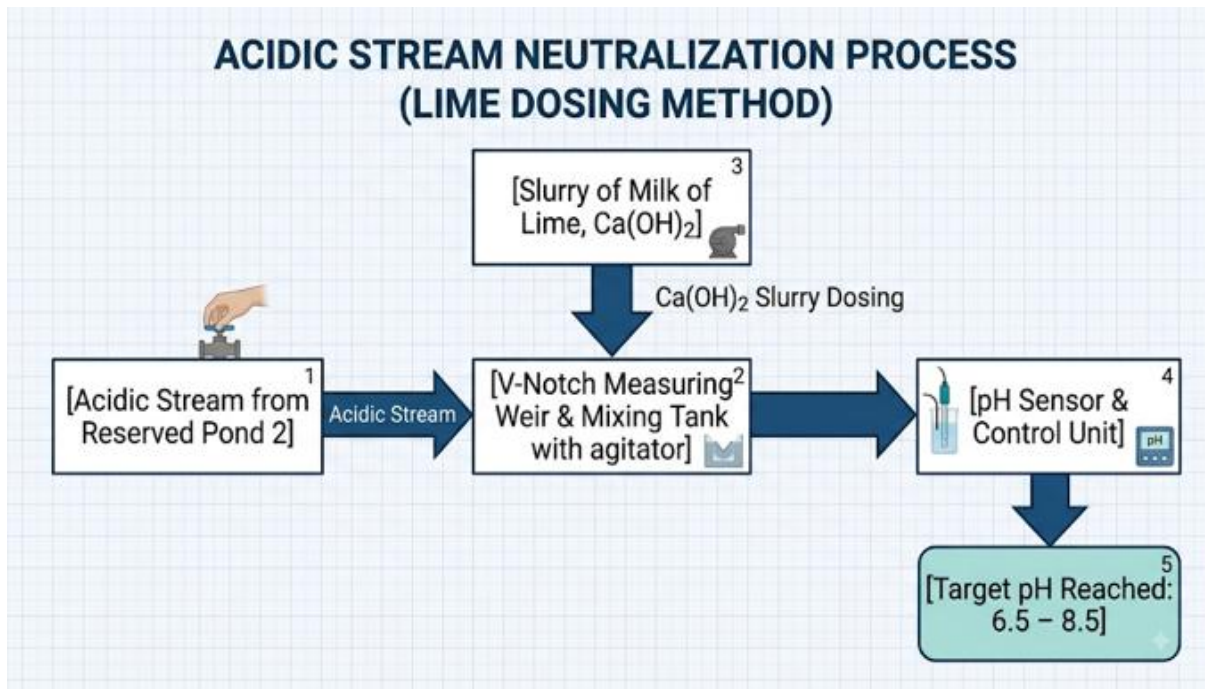
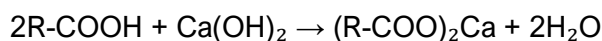


Figure 14: Lime Dosing Method

- **Lime Dosing Confirmation:** Milk of lime slurry (Ca(OH)<sub>2</sub>) is injected directly into the turbulent mixing zone at the weir.
- **Operational Goal:** Raw factory wastewater is inherently acidic (pH 4.5 – 5.5) due to sugar fermentation into organic acids. Acidic effluent suppresses biological bacterial activity and violates environmental guidelines. Lime dosing neutralizes organic acids and raises the stream pH into the strict NEMA Target Range (pH 6.5 – 8.5):



Furthermore, Ca<sup>2+</sup> ions react with residual phosphates and carbonates to form insoluble precipitates (Ca<sub>3</sub>(PO<sub>4</sub>)<sub>2</sub>, CaCO<sub>3</sub>), which act as weight agents to promote flocculation in the next treatment step.

### 6.3.2 SEDIMENTATION TANK SETTLING MECHANICS

Neutralized water from the weir enters a large concrete **Sedimentation Tank**.

- **Quiescent Settling:** The tank's large cross-sectional area drastically drops the fluid velocity, allowing suspended particulate matter, neutralized lime precipitates, and fine soil mud to settle to the bottom under gravity.
- **Sludge Scraping:** Settled sludge accumulates in a bottom collection pit where it is periodically de-sludged, while clarified supernatant water overflows over top weirs into Primary Pond 1.

### 6.3.3 PRIMARY PONDS: RETENTION TIME, VOLUME RATIO, AND SLUDGE DECANTING

The liquid enters Primary Pond 1 and progresses to Primary Pond 2. These large earthen basins provide extended Hydraulic Retention Time (HRT) ( $\approx 10\text{--}15$  days) necessary to reduce high biological loading.

- **Volume-to-Flow Ratio Index:** Hydraulic design ensures a high ratio of pond volume (V) to volumetric influent flow rate (Q):

$$\text{HRT (Days)} = V \text{ (m}^3\text{)} / Q \text{ (m}^3\text{/day)}$$

- **Mechanism:** Extended retention allows facultative microbes to digest colloidal organic matter. Heavy suspended organic sludges settle out to form a bottom benthic layer. Deep sludge undergoes anaerobic decanting and digestion, while upper water layers undergo surface aeration, stripping off residual BOD loading by up to 60%.

## 6.4 PHYTOREMEDIATION AND BIOLOGICAL POLISHING

### 6.4.1 MECHANICS OF PHYTOREMEDIATION VIA WATER HYACINTH (*EICHHORNIA CRASSIPES*)

Effluent flows from the primary ponds into **Secondary Pond 1** and **Secondary Pond 2**, which are engineered with surface mats of aquatic macrophyte vegetation—principally **Water Hyacinth (*Eichhornia crassipes*)**.

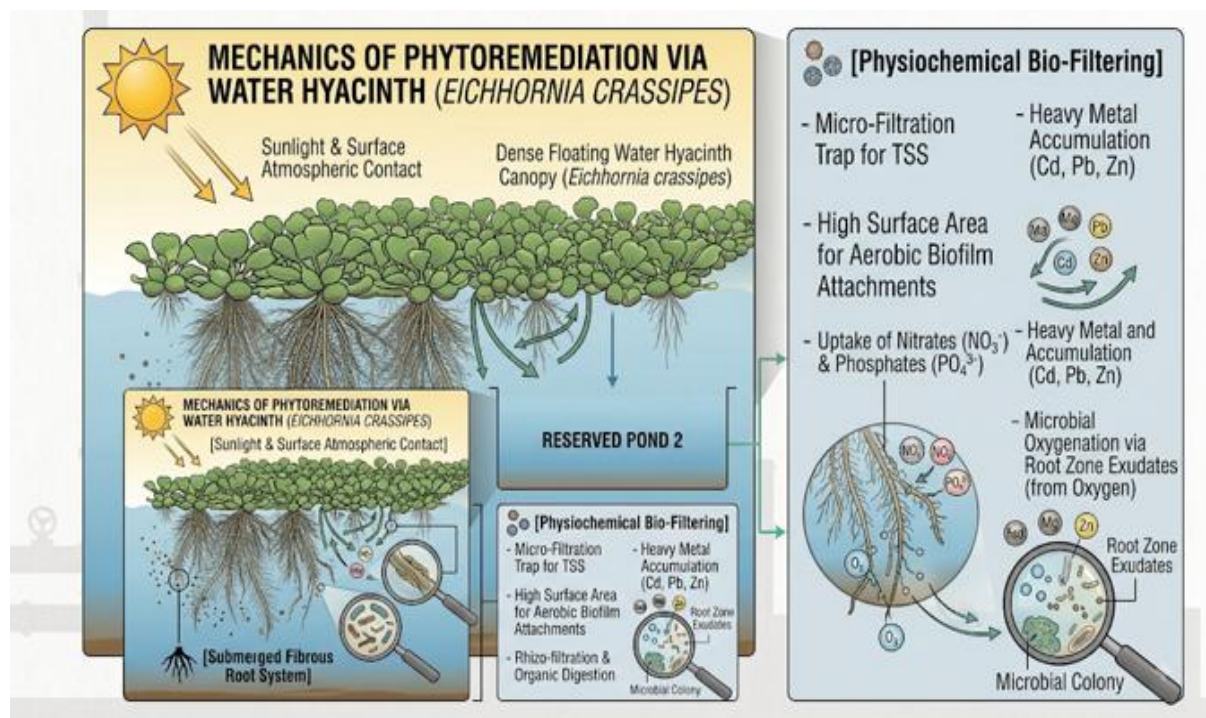


Figure 15: Mechanics of phytoremediation via water hyacinth

This biological purification system works via **Phytoremediation**:

1. **Nutrient Uptake (Rhizofiltration):** Water hyacinth plants rapidly absorb dissolved inorganic nutrients—specifically nitrates ( $\text{NO}_3^-$ ), ammonium ( $\text{NH}_4^+$ ), and orthophosphates ( $\text{PO}_4^{3-}$ ) generated from organic decomposition—utilizing them for vegetative biomass growth.
2. **Suspended Solids Trapping:** The dense, submerged, fibrous root network acts as a physical bio-filter screen that traps fine suspended solids (TSS) and colloidal flocs, clarifying turbidity.
3. **Microbial Biofilm Habitat:** Roots provide a massive surface area for attached-growth aerobic bacteria. Plants transfer atmospheric oxygen absorbed during photosynthesis down into the root zone (rhizosphere), creating oxygenated micro-zones that support nitrifying bacteria in an otherwise oxygen-depleted environment.
4. **Heavy Metal Accumulation:** Hyacinth roots adsorb trace heavy metals and complex mineral salts onto root cell walls via ion-exchange mechanisms, immobilizing toxic ions.

#### 6.4.2 FINAL STABILIZATION POND AND RIVER KUYWA DISCHARGE

The polished stream enters the Final Stabilization Pond. This shallow, clear-water basin allows final atmospheric re-aeration and sunlight UV exposure to destroy trace pathogens.

- **Visual & Parameter Status:** Effluent exiting the final pond is clear, odorless, low in turbidity, and rich in Dissolved Oxygen ( $\text{DO} > 5.0 \text{ mg/L}$ ).
- **Discharge:** The fully treated stream is discharged through a final monitoring flume directly into the River Kuywa, meeting all regulatory parameter thresholds established by the National Environment Management Authority (NEMA).

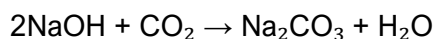
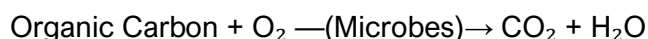
### 6.5 ETP LABORATORY ANALYTICAL PROCEDURES: BOD AND COD TESTING

Monitoring ETP efficiency requires continuous testing of physical and chemical parameters across influent, pond, and discharge stations.

#### 6.5.1 BIOCHEMICAL OXYGEN DEMAND ( $\text{BOD}_5$ ) VIA OXI TOP SYSTEM Industrial Importance & Principle

$\text{BOD}_5$  measures the dissolved oxygen consumed by aerobic microorganisms to biochemically oxidize organic matter in a water sample over a 5-day incubation period at  $20^\circ\text{C}$ .

Nzoia Sugar utilizes the automated OxyTop Respirometric System. Microorganisms in the sample consume dissolved oxygen and produce carbon dioxide (CO<sub>2</sub>). Sodium Hydroxide (NaOH) pellets placed in the bottle head-space absorb the generated CO<sub>2</sub> gas:



Absorption of CO<sub>2</sub> gas creates a pressure drop in the sealed bottle's headspace proportional to the volume of oxygen consumed. Electronic OxyTop sensor heads continuously record this pressure reduction.

### Reagents & Apparatus

- OxyTop Amber Glass Bottles (500 mL) with electronic measuring heads.
- Inductive magnetic stirring system and stir bars.
- Thermostatically controlled incubator cabinet (20°C ± 1°C).
- Sodium Hydroxide (NaOH) solid pellets.
- Allylthiourea (ATH) solution (Nitrification inhibitor to prevent ammonia oxidation).

### Analytical Procedure

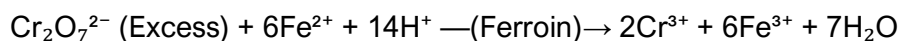
1. Determine the expected BOD range for the sample (e.g., Raw Influent vs. Final Outlet) to select the correct sample volume from OxyTop dosing tables (e.g., 435 mL for 0–40 mg/L range; 97 mL for 0–400 mg/L range).
2. Measure the selected sample volume using a graduated cylinder and transfer it into an OxyTop bottle containing a magnetic stir bar.
3. Add 5 drops of Allylthiourea (ATH) inhibitor to eliminate nitrogenous oxygen demand (NOD).
4. Insert a dry rubber sleeve into the bottle neck and place 2–3 solid NaOH pellets inside the sleeve (ensuring no pellets fall into the liquid sample).
5. Screw the electronic OxyTop measuring head tightly onto the bottle to form a hermetic seal, reset the digital display memory to zero, and place the bottle on the inductive stirrer rack inside the 20°C incubator.
6. Record daily respirometric pressure values for 5 consecutive days (120 hours).
7. Read the final 5-day value directly off the digital sensor screen in mg/L BOD<sub>5</sub> (or multiply by the bottle factor based on sample volume used).

## 6.5.2 CHEMICAL OXYGEN DEMAND (COD) VIA DICHROMATE DIGESTION & FAS TITRATION

### Industrial Importance & Principle

Chemical Oxygen Demand (COD) measures the total equivalent oxygen required to chemically oxidize both biodegradable and non-biodegradable organic compounds using a powerful oxidizing agent under strongly acidic thermal conditions.

Sample organics are oxidized by boiling with a known excess of Potassium Dichromate ( $\text{K}_2\text{Cr}_2\text{O}_7$ ) in high-concentration Sulfuric Acid ( $\text{H}_2\text{SO}_4$ ). The unreacted dichromate remaining after digestion is back-titrated against standard Ferrous Ammonium Sulfate (FAS) using Ferroin indicator:



### Reagents & Apparatus

- **OxyTop Amber Glass Bottles (500 mL):** With electronic measuring heads.
- Inductive magnetic stirring system and stir bars.
- Thermostatically controlled incubator cabinet ( $20^\circ\text{C} \pm 1^\circ\text{C}$ ).
- Sodium Hydroxide (NaOH) solid pellets.
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3. Insert a dry rubber sleeve into the bottle neck and place 2–3 solid NaOH pellets inside the sleeve (ensuring no pellets fall into the liquid sample).
4. Screw the electronic OxyTop measuring head tightly onto the bottle to form a hermetic seal, reset the digital display memory to zero, and place the bottle on the inductive stirrer rack inside the  $20^\circ\text{C}$  incubator.
5. Record daily respirometric pressure values for 5 consecutive days (120 hours).
6. Read the final 5-day value directly off the digital sensor screen in mg/L  $\text{BOD}_5$  (or multiply by the bottle factor based on sample volume used).

*Color change at end-point:* **Blue-green to sharp reddish-brown (wine-red).**

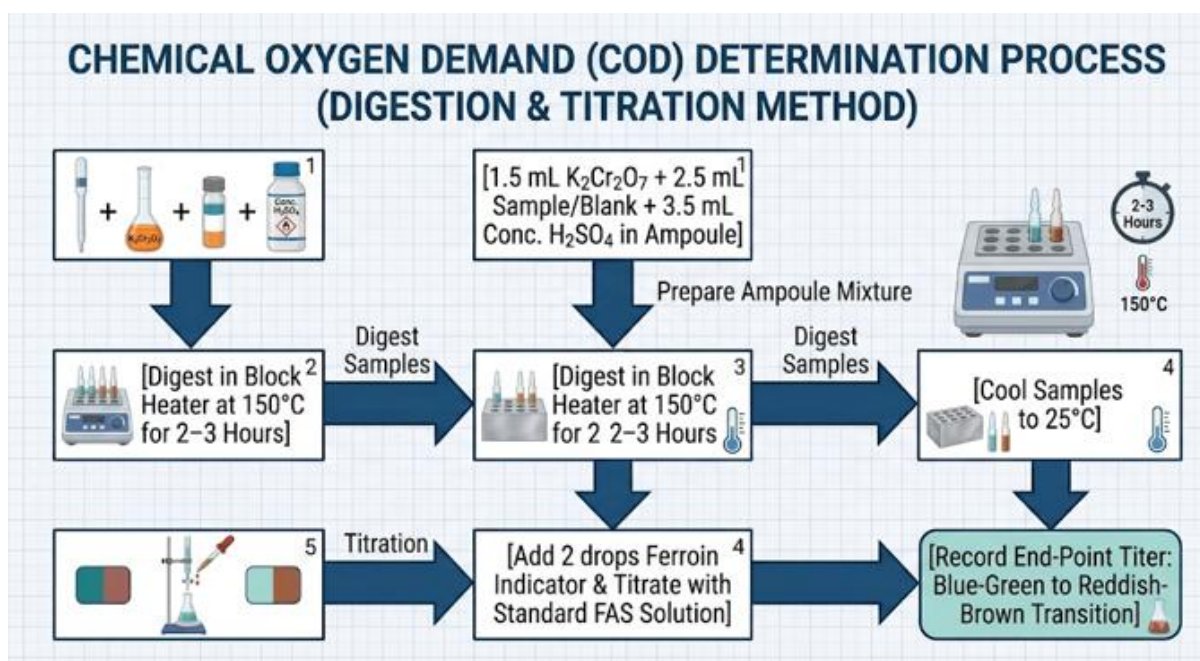


Figure 16: COD Determination Process

### Reagents & Standard Preparation

- **Digestion Solution (0.025 M  $K_2Cr_2O_7$ ):** Dissolve 1.2259 g of primary standard  $K_2Cr_2O_7$  (previously dried at 105°C) in 500 mL distilled water, add 100 mL concentrated  $H_2SO_4$  and 16.7 g  $HgSO_4$  (mercuric sulfate to mask chloride interference), and dilute to 1000 mL.
- **Sulfuric Acid Reagent:** Concentrated  $H_2SO_4$  containing Silver Sulfate ( $Ag_2SO_4$ ) catalyst (10 g  $Ag_2SO_4$  per 1 kg  $H_2SO_4$ ).
- **Standard Ferrous Ammonium Sulfate Titrant ( $\approx 0.05$  M FAS):** Dissolve 19.60 g of  $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$  in distilled water, add 20 mL conc.  $H_2SO_4$ , cool, and dilute to 1000 mL. Standardize daily against standard  $K_2Cr_2O_7$ .
- **Ferroin Indicator:** 1.485 g 1,10-phenanthroline monohydrate + 0.695 g  $FeSO_4 \cdot 7H_2O$  in 100 mL distilled water.

### Analytical Procedure

- **Sample Preparation & Dilution:** Dilute high-strength effluent samples with distilled water according to the expected COD range (e.g., Raw Influent diluted 1:10 or 1:20; Final Outlet tested neat/undiluted).
- **Ampoule/Reflux Loading:**

1. Pipette 1.50 mL of 0.025 M  $K_2Cr_2O_7$  digestion solution into clean digestion ampoules/tubes.



2. Add 2.50 mL of prepared sample (or diluted sample) into the tube.
  3. Prepare a Blank Solution by substituting 2.50 mL of distilled water for the sample.
  4. Carefully add 3.50 mL of concentrated Sulfuric Acid Reagent down the inner wall of each ampoule.
- **Thermal Digestion:** Cap the ampoules tightly, invert gently to mix, and place into a pre-heated thermo-reactor block heater at  $150^{\circ}\text{C} \pm 2^{\circ}\text{C}$  for 2 to 3 hours.
  - **Cooling & Titration:** Remove ampoules from the heater block and allow them to cool to room temperature ( $25^{\circ}\text{C}$ ). Transfer contents into a conical flask, add 2 drops of Ferroin indicator, and titrate with standard FAS titrant from a digital burette.
  - **End-Point Recognition:** Record the titrant volume when the solution transitions sharply from blue-green to deep reddish-brown.

### Calculations and Formulae

Calculate Chemical Oxygen Demand (COD) using the stoichiometric milliequivalent mass of oxygen (8000 mg  $\text{O}_2$ /equivalent):

$$\text{COD (mg O}_2\text{/L)} = [(V_{\text{blank}} - V_{\text{sample}}) \times N_{\text{FAS}} \times 8000 \times \text{DF}] / V_{\text{sample}} \text{ (mL)}$$

Where:

- **V<sub>blank</sub>** = Volume of FAS titrant used for the distilled water blank (mL).
- **V<sub>sample</sub>** = Volume of FAS titrant used for the effluent sample (mL).
- **N<sub>FAS</sub>** = Exact Normality / Molarity of FAS titrant (eq/L or mol/L).
- **8000** = Milliequivalent weight of oxygen (mg/eq).
- **DF** = Sample Dilution Factor (DF = 1 for undiluted sample).
- **V<sub>sample</sub> (mL)** = Volume of liquid sample added to tube (2.50 mL).

## 6.6 EFFLUENT QUALITY SPECIFICATIONS AND SHIFT ANALYTICAL DATA TABLE

The table below presents quality targets set by environmental regulations alongside shift analytical data across ETP sampling stations.

### Master ETP Shift Analytical Data and NEMA Regulatory Standards

| Effluent Quality Parameter and Test Unit | Ex-Factory Influent | Post-Lime Weir Sedimentation | Secondary Pond (Hyacinth Outlet) | Final Discharge (To River Kuywa) | NEMA Statutory Maximum Limit |
|------------------------------------------|---------------------|------------------------------|----------------------------------|----------------------------------|------------------------------|
|------------------------------------------|---------------------|------------------------------|----------------------------------|----------------------------------|------------------------------|

|                                          |            |            |           |           |              |
|------------------------------------------|------------|------------|-----------|-----------|--------------|
| pH (Glass Electrode)                     | 4.80       | 7.40       | 7.15      | 7.25      | 6.50 – 8.50  |
| Temperature (°C)                         | 38.5°C     | 29.0°C     | 25.5°C    | 24.0°C    | < 35.0°C     |
| Total Dissolved Solids (TDS)             | 1420 mg/L  | 1100 mg/L  | 620 mg/L  | 480 mg/L  | ≤ 1200 mg/L  |
| Total Suspended Solids (TSS)             | 850 mg/L   | 210 mg/L   | 45 mg/L   | 22 mg/L   | ≤ 30.0 mg/L  |
| Electrical Conductivity                  | 2180 µS/cm | 1680 µS/cm | 950 µS/cm | 740 µS/cm | ≤ 1500 µS/cm |
| Total Hardness (as CaCO <sub>3</sub> )   | 180 mg/L   | 340 mg/L   | 210 mg/L  | 160 mg/L  | N/A          |
| Total Alkalinity (as CaCO <sub>3</sub> ) | 45 mg/L    | 220 mg/L   | 180 mg/L  | 150 mg/L  | N/A          |
| Oil and grease                           | 125.0 mg/L | 12.0 mg/L  | 1.5 mg/L  | 0.2 mg/L  | ≤ 5.0 mg/L   |
| BOD <sub>5</sub> (OxyTop, 20°C)          | 1850 mg/L  | 680 mg/L   | 85 mg/L   | 18.5 mg/L | ≤ 30.0 mg/L  |
| COD (Dichromate Method)                  | 3920 mg/L  | 1410 mg/L  | 190 mg/L  | 42.0 mg/L | ≤ 50.0 mg/L  |
| Calculated COD: BOD Ratio                | 2.12       | 2.07       | 2.23      | 2.27      | N/A          |

Figure 3: Master ETP Shift Analytical Data and NEMA Regulatory Standards



# CHAPTER 7: THE QUALITY CONTROL LABORATORY STRUCTURE AND ROLE

## 7.1 LAB LAYOUT AND SECTIONS

Like most sugar-factory laboratories, the laboratory was organized into functional sections rather than one open room, so that different sample types and instruments did not interfere with one another. The main juice-process analysis, Masecuite and Molasses analysis benches has the polarimeter, Brix refractometer, and spindle, positioned away from heat sources and direct sunlight since both instruments are sensitive to temperature. A separate area was set aside for gravimetric/oven work (ash and moisture determination), since the muffle furnace and drying oven generate heat that could affect nearby optical instruments, Next bench is for Boiler water and Effluent analysis.

The special analysis is usually done in different section room.

The analysis includes:

### A. Daily Analysis

#### Sugar Material

- Sugar Pol
- Sugar Moisture
- Sugar Colour
- Water Insoluble Matter in Sugar
- Crystal Size of Sugar

### B.

|           |                                                                                                                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| Monday    | 1. Titrable Acidity in FEJ, CJ, and Syrup<br>2. Dextran in FEJ, MJ, CJ, and Syrup<br>3. Starch in FEJ, MJ, CJ, and Syrup |
| Tuesday   | 1. Reducing sugar in FEJ, MJ, CJ, and syrup<br>2. Total Reducing Sugar in Final Molasses Storage tank                    |
| Wednesday | 1. Preparation Index<br>2. Sucrose Analysis                                                                              |
| Thursday  | 1. Colour of CJ, Syrup, AL, and Melt<br>2. Fibre % cane                                                                  |
| Friday    | 1. Titrable acidity in FEJ, MJ, CJ, and Syrup<br>2. Dextran in FEJ, MJ, CJ, and Syrup                                    |

|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
|          | 3. Starch in FEJ, MJ, CJ, and Syrup                                                                |
| Saturday | 1. Reducing Sugar FEJ, MJ, CJ, and Syrup<br>2. Total Reducing Sugar in Final Molasses Storage tank |

Table 4: Special Daily Analysis timetable

### C. Quality check for chemical supplied to be used in process

- CaO in Lime
- Strength of Caustic soda
- Settling Test of juice Flocculant

### D. Calibration of Lab Equipment's on Weekly basis

This special Analysis testing was carried out at a distinct bench, this side, partly for convenience and partly i.e. because effluent samples are handled separately from food-contact sugar samples for hygiene reasons.

Standard safety features expected in a lab of this kind include a fume hood or extraction point for work involving concentrated acids (e.g. sulphated ash, COD digestion), an eyewash station, a fire extinguisher, and clearly labelled reagent storage.

## 7.2 EQUIPMENT AND INSTRUMENTATION

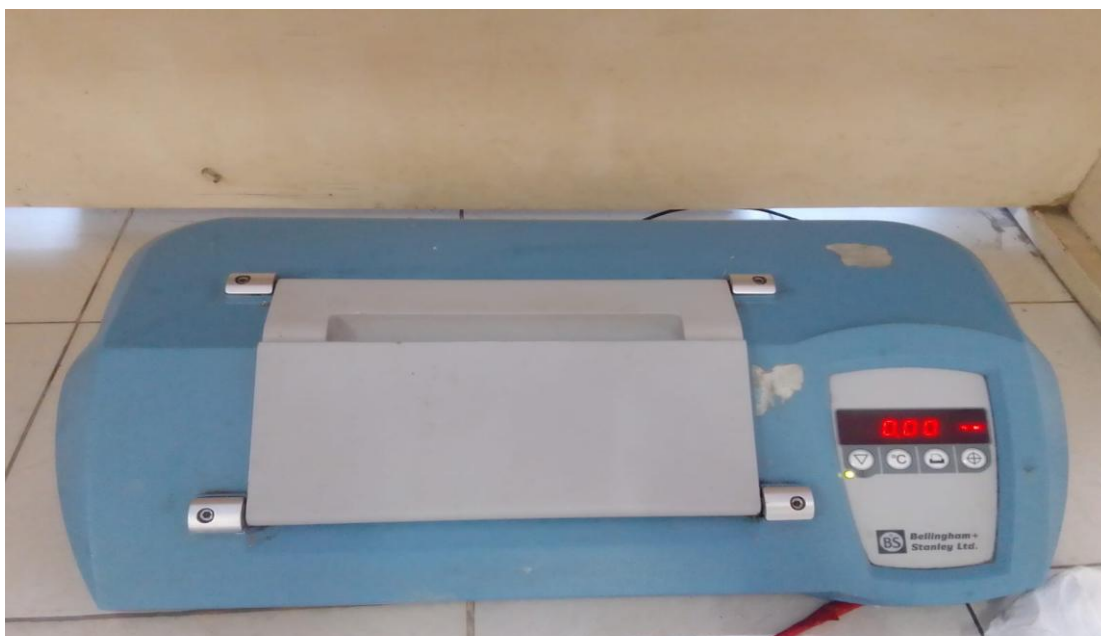
The laboratory was equipped with instruments covering both sugar-process analysis and environmental/effluent testing:

### 7.2.1 POLARIMETER

#### Principle

A polarimeter operates on the principle of optical rotation, where optically active substances rotate the plane of polarized light passing through them. The angle of rotation depends on the concentration and nature of the substance, as well as the path length of the sample.

It is used to determine the Pol (apparent sucrose content) of juice, massecuite, molasses, syrup, and sugar samples by measuring their optical rotation.



## Principle

A Brix refractometer operates on the principle of refraction of light, where the refractive index of a solution changes according to the concentration of dissolved substances. The measured refractive index is converted into a Brix value.

It is used for rapid determination of Brix, which represents the percentage of soluble solids in juice, syrup, and other sugar solutions.



Principle

A Brix spindle operates based on the principle of buoyancy, where an object immersed in a liquid experiences an upward force related to the density of the liquid. The depth to which the spindle floats is used to determine the specific gravity and corresponding Brix value.

It is used as a manual method for determining the Brix of sugar solutions and can also be used to cross-check refractometer results.



Figure 19: Hydromete

## 7.2.4 DRYING OVEN

### Principle

A drying oven operates by using controlled heat to remove moisture or other volatile substances from a sample. The difference between the initial and final mass of the sample after drying is used to determine the moisture content or loss on drying.

It is used for moisture determination and loss-on-drying analysis by heating weighed samples at a specified temperature until a constant mass is obtained.



Figure 20: Drying Oven

## 7.2.5 OXITOP BOX (BOD APPARATUS)

### Principle

The OxiTop system determines Biochemical Oxygen Demand (BOD) based on the consumption of dissolved oxygen by microorganisms as they biologically decompose organic matter in a sample. The resulting pressure change in the sealed bottle is measured and related to the oxygen consumed.

It is used to determine the BOD of factory effluent and wastewater samples, commonly over a five-day incubation period. The result indicates the amount of biodegradable organic matter present in the wastewater.



Figure 21: OxiTop

## 7.2.6 COD REACTOR

### Principle

The COD reactor operates on the principle of chemical oxidation, in which organic matter in a wastewater sample is oxidized using a strong oxidizing agent under controlled heating conditions. The amount of oxygen required for this oxidation is expressed as the Chemical Oxygen Demand (COD).

It is used to digest wastewater or effluent samples during COD analysis, providing a measure of the total oxygen-equivalent of chemically oxidizable substances present in the sample. The test is generally faster than BOD analysis.



Figure 22: COD Reactor

## 7.2.7 pH Meter

### Principle

The pH meter operates by measuring the electrical potential difference between a glass electrode and a reference electrode when immersed in a solution. The potential difference is related to the hydrogen ion activity of the solution.

It is used to determine the acidity or alkalinity of samples and solutions by measuring their pH.



Figure 23: pH meter

## 7.2.8 Conductivity Meter

### Principle

The conductivity meter operates by measuring the ability of a solution to conduct an electrical current. Electrical conductivity depends mainly on the concentration and mobility of ions present in the solution.

It is used to determine the ionic content and assess the quality or purity of water and other solutions.





Figure 24: Conductivity meter

## 7.2.9 Spectrophotometer

### Principle

The spectrophotometer operates by measuring the amount of light absorbed by a sample at a selected wavelength. The absorbance is related to the concentration of the substance according to the Beer-Lambert law.

It is used for qualitative and quantitative analysis, particularly for determining the concentration of substances in solution.

## 7.2.10 Analytical Balance

### Principle

The analytical balance uses an electromagnetic force-compensation system to accurately determine the mass of a sample. The force required to balance the sample is converted into a precise mass reading.

It is used for accurately weighing chemicals and samples during preparation of solutions, standards, and laboratory analyses.





Figure 25: Analytical balance

## **CHAPTER 8: ANALYTICAL PROCEDURES AND METHODS PERFORMED**

This chapter documents the analytical procedures actually followed in the Nzoia Sugar Company laboratory during the attachment, based on the laboratory's own standard operating procedures. Sample collection points and methods are described first, followed by each analytical determination in the sequence generally used in the lab: solids (Brix), Pol, purity, reducing sugars, ash, moisture, pH, colour, and the specialised process-control and environmental tests carried out on a daily, weekly, and as-needed basis.

### **8.1 SAMPLE COLLECTION AND PRESERVATION**

Representative sampling is critical to reliable analysis, since a single unrepresentative sample can give a misleading picture of an entire batch. The laboratory follows defined sampling points and methods for each stream in the process:

#### **First Expressed Juice (FEJ)**

Hand-sampled with a spoon across the full width of the feed rollers on the feed side of the first mill, at three 15-minute intervals to build a composite sample.

#### **Mixed Juice (MJ)**

Caught every 15 minutes via a tap on the delivery pipe from the mixed juice tank; the composite is preserved with mercuric chloride for one hour before laboratory analysis.

#### **Last Expressed Juice (LEJ)**

Hand-sampled with a spoon across the discharge roller, composited at 15-minute intervals from the discharge side of the mill.

#### **Clarified/Clear Juice (CJ)**

Sampled from the pipe leading to the evaporators via a tap/cork.

#### **Raw Syrup**

Sampled through a tap on the delivery pipe linking the evaporators to the raw syrup receiving tanks.

#### **Treated Syrup**

Sampled as it leaves the pipe linking the evaporators to the raw syrup receiving tanks, after sulphitation/treatment.

#### **Bagasse**

Spot-sampled by hand across the full width of the bagasse overflow from the last mill discharge every 15 minutes, composited over one hour, and preserved with an ammonia-chloroform solution to prevent sucrose deterioration.

### **Molasses**

Sampled by hand through a tap on the pipe leading to the molasses storage tank.

### **Massecuite (A, B, C)**

Sampled individually per strike from the pan sampling cork or proof stick before the massecuite is struck (discharged from the pan).

### **Magma**

Sampled after the mingler, before the material proceeds to the pans.

### **Filter Cake**

Sampled across the full length of the vacuum filter drum, composited in a closed container at 15-minute intervals (three times), and taken for immediate analysis.

### **Sugar**

Sampled after the bin, before bagging, using a dry sampling container; a 24-hour composite is prepared from samples taken across the three shifts.

### **Effluent**

Raw effluent is sampled hourly after chemical dosing; treated effluent is sampled once daily.

### **Prepared Cane (for direct analysis)**

Sampled across the conveyor after the shredder at 15-minute intervals, with the composite kept sealed in a sample bucket.

## **8.2 Determination of Solids (Brix)**

Brix (total dissolved solids) is determined using a digital refractometer across all major process streams.

### **8.2.1 JUICES (FEJ, MJ, CLARIFIED JUICE, LEJ, FILTRATE)**

The refractometer prism plates are opened and cleaned with distilled water and tissue. A few drops of the strained sample are placed on the lower plate and the prism closed. The eyepiece is focused to obtain a sharp, straight interface between the light and dark fields, which is brought to the cross-hair, and the Brix and temperature are read off (digital units display the result directly). The instrument is cleaned between samples.

### **8.2.2 Syrups**

One or two drops of syrup are placed on the lower prism, the upper prism closed, and the same focusing procedure followed to obtain the Brix and temperature reading.

### 8.2.3 MASSECUITES (A, B, C) AND MOLASSES (A, B, C)

The massecuite or molasses sample is diluted 1:10 with distilled water in a beaker and stirred until fully dissolved, since these products are too viscous/dark to read directly. Brix is then determined as for juice, and the reading is multiplied by a factor of 10 to correct for the dilution.

## 8.3 DETERMINATION OF POL

Pol (apparent sucrose content) is determined using a saccharimeter/polarimeter, with clarification steps that vary by sample type.

### 8.3.1 JUICES (FEJ, MJ, CLEAR JUICE, FILTRATE, AND LEJ)

200 mL of sample is measured into a 250 mL beaker and about 1 g of dry lead acetate is added and stirred in vigorously to clarify the solution, which is left to stand briefly. The mixture is filtered, discarding the first 15 mL of filtrate (which is not returned to the filter). Where the filtrate is cloudy, a drop of glacial acetic acid is added to clear it. A 200 mm saccharimeter tube is rinsed three times with filtrate, then filled without trapping air bubbles, and read for zero setting before the pol is read. The Brix (uncorrected for temperature) together with the corrected pol reading is used to obtain the final pol via a Schmitz conversion table for undiluted juice.

| Sample | pH  | Brix       | PRD   | OC Brix | Pol%  | Purity (%) |
|--------|-----|------------|-------|---------|-------|------------|
| FEJ    | 5.4 | 15.8(24°)  | 53.42 | 16.05   | 13.07 | 81.4       |
| LEJ    | 5.7 | 1.8        | 5.4   | 1.8     | 1.3   | 72.2       |
| MJ     | 5.3 | 11.3(30°C) | 35.96 | 11.96   | 8.95  | 74.9       |
| CJ     | 6.4 | 10.0(29°C) | 34.68 | 10.58   | 8.92  | 84.3       |

Table 5: Juices analysis examples

### 8.3.2 SYRUPS

26 g of syrup is weighed into a 100 mL volumetric flask, made up to the mark with distilled water, shaken, and poured into a 250 mL beaker for filtration as with juices. The pol is read in a 200 mm tube using the polarimeter, and the reading is taken directly as the pol % of the product.

### 8.3.3 MASSECUITES (A, B, C) AND MOLASSES (A, B, AND FINAL)

26 g of the 1:10 diluted solution already prepared for Brix determination is transferred to a 100 mL volumetric flask and made to the mark with distilled water. The solution is mixed, poured into a 250 mL beaker, and clarified with dry lead subacetate — 6 g for C-grade, 4 g for B-grade, and 2 g for A-grade massecuites/molasses — then stirred and filtered. The pol is read in a 200 mm tube and the reading multiplied by 2 to obtain the pol percentage.

### 8.3.4 SUGAR

26 g of sugar is weighed into a 100 mL volumetric flask, dissolved in about 70 mL of distilled water, treated with 5 mL of wet lead subacetate, and made up to the mark. After filtering and noting the filtrate temperature, the sample is polarized in a 200 mm tube. A temperature correction is applied using  $\text{Pol}_{20} = \text{pol reading} \div [1 - 0.000184 \times (\text{sample temperature} - 20)]$ .



Figure 26: carrying out sugar pol, moisture analysis.

|                |         |
|----------------|---------|
| Initial weight | 33.8244 |
| Final weight   | 33.8175 |
| Difference     | 0.0069  |
| Pol            | 99.99%  |

Table 6: Sugar analysis example

## 8.4 PURITY CALCULATION

Purity (%) is calculated, not directly measured, as  $\text{Purity} = (\text{Pol} \div \text{Brix}) \times 100$ , using the Brix and pol values obtained for the same sample. Purity is tracked at each process stage (juice, syrup, massecuite/molasses grades) as a key indicator of boiling house and clarification efficiency

## 8.5 DETERMINATION OF REDUCING SUGARS (RS%)

Reducing sugar is determined titrimetrically against Fehling's solutions, using methylene blue as the end-point indicator.

### 8.5.1 JUICE

100 mL of screened juice is pipetted into a 200 mL volumetric flask and made to volume with distilled water (diluted further to 50 or 25 mL of juice if the reducing sugar percentage exceeds 0.65%). 5 mL each of Fehling's solutions A and B are pipetted into a 250 mL conical flask containing glass beads. A 50 mL burette is rinsed and filled with the diluted juice. 15 mL of the diluted juice is run into the flask, boiled for 2 minutes, and 3–4 drops of 2% methylene blue indicator are added directly to the boiling liquid while titration continues. The titration ends when the indicator's blue colour changes fully to orange, with boiling maintained throughout. A minimum of three titrations (one rough, two accurate) is performed.

### 8.5.2 SYRUP

20 g of syrup is weighed into a 200 mL volumetric flask and diluted to the mark with distilled water, then titrated against 10 mL each of Fehling's solutions A and B, following the same indicator and end-point procedure as for juice, noting the Brix, pol, and titre value of the sample.

Gram sucrose per 100 mL =  $(\text{Pol \% of juice} \times \text{volume of sample taken}) \div \text{dilution volume (200)}$ . This figure is used to read the corresponding mg RS/100 mL from standard reference tables through interpolation, from which RS% and the RS/Brix ratio are calculated.

**Target Reference Standards across Factory Juice Streams.**

| Parameter           | First Expressed Juice (FEJ) | Mixed Juice (MJ) | Clear Juice (CJ) | Raw Syrup     |
|---------------------|-----------------------------|------------------|------------------|---------------|
| pH Target           | 5.20 – 5.60                 | 5.00 - 5.40      | 6.80 – 7.20      | 6.20 – 6.60   |
| Observed Brix (°Bx) | 18.00 – 21.00               | 14.00 – 16.00    | 15.00 – 17.00    | 60.00 – 65.00 |
| Pol Target (%)      | 15.00 – 18.00               | 11.50 – 14.00    | 12.50 – 15.00    | 50.00 – 56.00 |
| Target Purity (%)   | 82.00 – 86.00               | 80.00 – 84.00    | 82.00 – 85.00    | 82.00 – 85.00 |

Table 7: Target Reference Standards across Factory Juice Streams.

## 8.6 ANALYTICAL PROCEDURES AND SAMPLE CALCULATION DATA TABLE

To demonstrate the application of these formulas, the table below contains laboratory bench data collected during a shift rotation, showing the step-by-step mathematical conversion from raw spindle readings to final purity metrics.

| Analytical Parameters and Calculation | First Expressed Juice (FEJ) | Mixed Juice (MJ) | Clear Juice (CJ) | Raw Syrup (1:1 Dilution)           |
|---------------------------------------|-----------------------------|------------------|------------------|------------------------------------|
| Sample Volume Analysed                | 250 mL                      | 250 mL           | 250 mL           | 125g Juice + 125g H <sub>2</sub> O |
| Raw Spindle Brix Reading              | 18.3°BX                     | 11.6°BX          | 11.75°BX         | 6.5°BX                             |
| Observed Juice Temp (\$T\$)           | 26°C                        | 31°C             | 29°C             | 24°C                               |

|                                              |                                    |                                 |                                 |                                 |
|----------------------------------------------|------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| <b>Temp Correction Factor (+)</b>            | +0.48°Bx                           | +0.52°Bx                        | +0.36°Bx                        | +0.32°Bx                        |
| <b>Analytical Parameters and Calculation</b> | <b>First Expressed Juice (FEJ)</b> | <b>Mixed Juice (MJ)</b>         | <b>Clear Juice (CJ)</b>         | <b>Raw Syrup (1:1 Dilution)</b> |
| <b>Corrected Brix (°Bx<sub>20</sub>)</b>     | 18.75°BX                           | 12.46°BX                        | 12.33°BX                        | 6.5°BX                          |
| <b>Pol Reading (°Z, 200mm Tube)</b>          | 68.32°Z                            | 43.28°Z                         | 44.04°Z                         | 22.38°Z                         |
| <b>Schmitz Factor (from Table)</b>           | 0.2525                             | 0.2578                          | 0.2470                          | 0.2562                          |
| <b>Calculated Pol %</b>                      | 17.25                              | 11.16                           | 10.88                           | 5.73                            |
| <b>Formula for Purity</b>                    | $(17.25 \div 18.75) \times 100$    | $(11.16 \div 12.46) \times 100$ | $(10.88 \div 12.33) \times 100$ | $(5.73 \div 6.5) \times 100$    |
| <b>Apparent Purity (%)</b>                   | 92.00%                             | 89.55%                          | 88.22%                          | 88.82%                          |



|                   |      |      |      |      |
|-------------------|------|------|------|------|
| <b>pH Reading</b> | 5.35 | 5.18 | 7.10 | 6.42 |
|-------------------|------|------|------|------|

Table 8: sample calculation data table

## 8.7 DETERMINATION OF SUSPENDED SOLIDS IN MIXED JUICE

A catch sample (approximately 50 g) of mixed juice is weighed in a pre-weighed dry 1000 mL beaker to obtain the weight of juice by difference. 1.4 g of kieselguhr (diatomaceous filter aid) is added and the mixture boiled until bubbling starts, then filtered under vacuum through a pre-weighed filter paper. The residue and filter paper are dried in an oven at 105°C for 4 hours, cooled in a desiccator for 30 minutes, and reweighed to obtain the suspended solids by difference.

## 8.8 DETERMINATION OF SUGAR (SULPHATED) ASH

A weighed sugar sample in a platinum dish is moistened with concentrated sulphuric acid (or 10% sulphuric acid) and gently heated from above and below to char the sample without overflowing. The carbonized sample is placed in a muffle furnace at 650°C for three hours, removed, cooled, re-moistened with a drop of sulphuric acid, and returned to the furnace at 650°C for a further hour before final cooling in a desiccator. Sulphated ash (%) = (weight of ash × 100) ÷ weight of sample.

## 8.9 WATER INSOLUBLE MATTER IN SUGAR

Membrane filters (8 micrometre pore size) are pre-boiled in distilled water, dried at 105°C for 1 hour, cooled in a desiccator, and weighed (W). 100 g of sugar is dissolved in 700 mL of boiling distilled water and filtered under vacuum through the prepared membranes. The residue is dried at 105°C for 30 minutes, cooled, and weighed (X). Water insoluble matter =  $(X - W) \times 10,000$  mg/kg.

## 8.10 PH DETERMINATION

pH is determined electrometrically on juices, syrup, boiler water, and condensates, following calibration of the pH meter against standard buffer solutions before use.

## 8.11 MOISTURE AND POL IN FILTER CAKE

Moisture in filter cake follows the same drying/weighing procedure used for bagasse moisture (oven-drying to constant weight). For pol, 25 g of filter cake is weighed into a tared 250 mL beaker, made up to 200 mL with water, treated with 0.5 g of dry lead acetate, stirred, and left to stand briefly. The mixture is filtered (discarding the first 25 mL) and the filtrate polarized in a 200 mm tube, with the pol reading multiplied by 2.

## 8.12 QUALITATIVE TEST FOR SUGAR TRACES (IN WATER)

A suspected water sample is cooled in a test tube, treated with 5 drops of 2% 1-naphthol solution and shaken, then concentrated sulphuric acid is added slowly down the side of the tube without mixing. A purple ring at the acid-sample interface confirms the presence of sugar traces. The 1-naphthol reagent (prepared in 95%

alcohol) darkens on exposure to light and is stored in an amber bottle, with a fresh batch prepared roughly monthly.

### **8.13 SUCROSE AND ASH IN FINAL MOLASSES**

4 g of final molasses is weighed into a 250 mL volumetric flask, made to the mark with distilled water, and warmed to 40°C in a water bath. The solution is transferred to a beaker, treated with 1 g potassium oxalate and 4 g kieselguhr, and filtered through Whatman No. 91 filter paper.

#### **8.13.1 TOTAL INVERT**

62.5 mL of filtrate is pipetted into a 250 mL volumetric flask with 80 mL of distilled water, boiled to 60°C, treated with 10 mL of 6.34 N hydrochloric acid, and left standing for 30 minutes. Three drops of phenolphthalein indicator are added, followed by 4 N sodium hydroxide until the sample turns pink, before cooling and making to the mark. The sample is titrated against Fehling's A and B solutions (10 mL each) using methylene blue as indicator.

#### **8.13.2 REDUCING SUGARS**

The original filtrate is titrated directly against 10 mL each of Fehling's solutions A and B, again using methylene blue as indicator. Total invert and RS% are obtained from standard reference tables (Queensland Laboratory Manual Table IV) using the appropriate sucrose columns, and  $\text{Sucrose (\%)} = (\text{Total Invert} - \text{RS}) \times 0.95$ .

#### **8.13.3 ASH**

10 g of molasses (diluted 1:1) is weighed into a platinum dish, treated with about 10 mL of concentrated sulphuric acid, and heated gently until charred, then further heated/furnaced at 550–600°C for two hours until a pinkish-grey ash results. The ash is moistened with a few drops of sulphuric acid, reheated at 650°C for 30 minutes, cooled in a desiccator, and weighed.  $\text{Sulphated ash (\%)} = (\text{weight of ash} \times 100) \div \text{weight of sample}$ .

### **8.14 BIOCHEMICAL OXYGEN DEMAND (BOD) — PERMANGANATE OXIDATION METHOD**

The sample is first sealed in an OxiTop bottle and incubated in the dark at a controlled temperature (typically 20°C) for 5 days (the standard BOD<sub>5</sub> period). At the end of incubation, rather than reading the result directly off the OxiTop head, the routine BOD determination at Nzoia is completed using a potassium permanganate oxidation/titration method referenced against a calibration gradient.

#### **8.14.1 BLANK**

90 mL of distilled water is placed in a 250 mL conical flask with 10 mL of N potassium permanganate and 10 mL of 25% sulphuric acid, then immersed in a boiling water bath for 30 minutes, removed, and immediately treated with 12.5 mL of 0.8 N potassium oxalate. The mixture is titrated while hot with 0.1 N potassium permanganate and the titre noted.

### 8.4.2 SAMPLE

5 mL of sample is made up to 90 mL with distilled water in a 250 mL conical flask, treated with 10 mL of 1 N potassium permanganate and 10 mL of 25% sulphuric acid, boiled for 30 minutes in a water bath, then immediately treated with 12.5 mL of 0.8 N potassium oxalate and titrated hot with 0.1 N potassium permanganate.

BOD = [(Sample titre – Blank titre) × 0.833] ÷ 0.0077 (graph gradient), giving a result in ppm. For example, with a blank titre of 2.1 and sample titre of 7.3: BOD = (7.3 – 2.1) × 0.833 ÷ 0.0077.

### 8.15 DETERMINATION OF POL % CANE

250 g of prepared cane is weighed into a stainless-steel disintegrator bucket with 2000 mL of clean cold water, sealed, and disintegrated for 15 minutes. The liquid is decanted and the juice filtered, clarified with adequate lead subacetate, and the pol read in a 400 mm tube. The pol reading is multiplied by a moisture-percent correction factor to give pol % cane.

### 8.16 DETERMINATION OF FIBRE % CANE

500 g of prepared cane is weighed into a stainless-steel bucket, sealed, and ground for 15 minutes. 250 g of the resulting ground fibre is transferred into a pre-weighed dry cloth bag, tied tightly, and washed in hot then cold water until free of sugar traces (checked with the naphthol test in Section 5.11). The fibrous residue is oven-dried at 105°C for 24 hours, cooled in a desiccator for at least 30 minutes, and weighed.

Fibre % cane = (weight of dry fibre × 100) ÷ 250.

### 8.17 DETERMINATION OF PREPARATION INDEX (PI)

Preparation index measures how effectively the cane preparation equipment (knives/shredder) has ruptured cane cells before milling, comparing readily extractable ('rapi') pol against total ('leachable') pol.

#### 8.17.1 RAPI POL

250 g of prepared cane is weighed into a stainless bucket with 2000 mL of clean cold water and disintegrated for 15 minutes. The liquid is decanted, the juice filtered, clarified with lead subacetate, and the pol read in a 400 mm tube — this is the rapi pol.

#### 8.17.2 LEACHABLE POL

500 g of prepared cane is weighed into a 10-litre plastic container with 4000 mL of clean cold water and stirred manually for 30 minutes. The extract is strained, filtered with adequate lead subacetate, and the pol read in a 400 mm tube — this is the leachable pol.

Preparation Index (%) = (leachable pol × 100) ÷ rapi pol.

### 8.18 BOILER WATER ANALYSIS

Boiler feed water is monitored regularly to prevent scaling and corrosion in the boiler, with samples cooled to room temperature before analysis.

### 8.18.1 TOTAL, CALCIUM, AND MAGNESIUM HARDNESS

100 mL of water sample is treated with 1 mL of ammonium buffer and 3 drops of Eriochrome Black T indicator (a purple colour indicates hardness present; blue indicates none), then titrated with 0.01 M EDTA to a colour change from purple to faint blue for total hardness (Total Hardness = titre  $\times$  10, as ppm CaCO<sub>3</sub>). Calcium hardness is determined similarly on a fresh 100 mL sample treated with 4 mL of 1 N sodium hydroxide instead of buffer, titrated the same way. Magnesium hardness is obtained by subtracting calcium hardness from total hardness.

### 8.18.2 ALKALINITIES

Phenolphthalein alkalinity (P) is measured by titrating 100 mL of sample with 0.02 N hydrochloric acid after adding phenolphthalein indicator ( $P = \text{titre} \times 10 \text{ ppm}$ ). Hydroxide alkalinity (O) uses a similar titration after adding barium chloride and phenolphthalein ( $O = \text{titre} \times 10 \text{ ppm}$ ). Total alkalinity (M) is titrated with methyl orange indicator ( $M = \text{titre} \times 10 \text{ ppm}$ , or  $M = 2P - O$ ).

### 8.18.3 TOTAL DISSOLVED SOLIDS (TDS) AND CHLORIDE

TDS is estimated from total alkalinity ( $\text{TDS} = M \times 10 \div 5$ ) or measured directly with a TDS meter. Chloride is determined by titrating 50 mL of sample (with phenolphthalein and dilute sulphuric acid to remove alkalinity, then potassium chromate indicator) against 0.0171 M silver nitrate to a brick-red end point; Chloride as NaCl (mg/L) = titre  $\times$  20.

## 8.19 DETERMINATION OF AVAILABLE CAO IN LIME

Two alternative methods are used to assess lime quality before use in juice clarification.

### 8.19.1 SUCROSE METHOD

1.6 g of lime powder is treated with 2 mL of ethyl alcohol and 100 mL of 10% sucrose solution, shaken for 30 minutes, and filtered (discarding the first 5 mL). 50 mL of filtrate is titrated against 1 N HCl using methyl orange indicator. % Available CaO =  $(\text{titre} \times N \text{ HCl} \times 5.6) \div 1.6$ .

### 8.19.2 PHENOL METHOD

2.5 g of powdered lime is transferred into a 250 mL volumetric flask with 5% aqueous phenol solution, made to the mark, shaken for at least an hour, and filtered (discarding the first 50 mL). % Available CaO =  $(\text{mL of N HCl used} \times 5.6) \div \text{weight of lime}$ .

## 8.20 DETERMINATION OF EXTRANEEOUS MATTER IN CANE

Six cane samples delivered to the factory are cut into pieces and weighed (M), then trash is removed and weighed (N). The samples are washed to remove adhering soil, and the washings collected and treated with 3 g of kieselguhr, boiled for 2 minutes, and filtered through a pre-weighed dry filter paper under vacuum. The residue and filter paper are oven-dried at 105°C for 24 hours, cooled in a desiccator, and

weighed (Z). Extraneous matter % cane =  $[(N + Y) \times 100] \div M$ , where Y is the residue weight (Z minus filter paper weight).

### **8.21 SLURRY PREPARATION (FOR PAN SEEDING)**

3 kg of sugar is disintegrated in a ball mill with 5 litres of methylated spirit for 24 hours to form a seed slurry, typically at a 1:2 ratio for varying volumes; 1.5 litres is issued for a single graining to give three C-grade strikes of 350 hL each. Where final molasses purity is observed to be rising, the sugar quantity is increased and the number of C-sugar strikes reduced to improve exhaustion (e.g. 2.8 kg sugar in 5 litres of spirit, run 2.5 hours, issuing 2.75 litres for a single graining to give two C-strikes).

### **8.22 PRIMARY EXTRACTION (BOILING HOUSE EFFICIENCY)**

Primary extraction, a key measure of milling efficiency, is assessed by two complementary methods.

#### **7.22.1 POL % METHOD**

Pol in prepared cane, moisture and fibre; first-mill discharge juice Brix, pol, and purity; and first-mill bagasse pol and moisture are all determined. Brix of mill 1 bagasse =  $(\text{mill 1 bagasse pol} \times 100) \div \text{mill 1 discharge juice purity}$ , and fibre % bagasse =  $100 - \text{brix} - \text{moisture}$ . From these, fibre % cane, pol in mill 1 bagasse % cane, and pol in mill 1 juice % cane are derived, giving Primary Extraction (%) =  $(\text{pol in mill 1 juice \% cane} \times 100) \div \text{pol \% cane}$ .

#### **8.22.2 JUICE METHOD**

Using fibre % cane, first-mill discharge juice Brix/pol/purity, and first-mill bagasse pol/moisture, brix % bagasse and fibre % mill 1 bagasse are calculated, from which mill 1 bagasse % cane, juice extracted in mill 1, and absolute juice % cane are derived, giving Primary Extraction (%) =  $(\text{juice extracted in mill 1} \times 100) \div \text{absolute juice \% cane}$ .

### **8.23 TOTAL SUSPENDED SOLIDS (TSS) IN WATER**

TSS is determined either by spectrophotometer (set at 810 nm, calibrated to zero absorbance/100% transmittance with a distilled water blank, then read directly against the sample; TSS =  $\text{sample reading} \times 1000 \text{ ppm}$ ) or gravimetrically (a pre-weighed glass microfibre filter is used to filter 100 mL of sample under vacuum, dried at 105°C for 1 hour, cooled, and reweighed; TSS =  $\text{mass of dry suspended solids} \times 10^6 \div \text{volume of sample}$ ).

### **8.24 COLOUR ANALYSIS (SYRUP AND SUGAR SOLUTIONS)**

Syrup samples are diluted to  $5.0 \pm 1.0$  Brix and adjusted to  $\text{pH } 7.00 \pm 0.05$  before filtration through a 0.45 micrometre membrane, and read at 420 nm against a distilled-water-zeroed spectrophotometer. For white sugar solutions, both unfiltered and filtered (0.45 micrometre membrane) absorbance are read at 420 nm, allowing colour and turbidity to be calculated separately: Colour =  $(\text{Absorbance} \times 1000) \div (\text{concentration in g/cm}^3 \times \text{cell diameter})$ , with concentration read from a reference solids/Brix table; Turbidity = Unfiltered colour – Filtered colour, since filtered colour represents the true colour of the sugar.

### **8.25 TURBIDITY DETERMINATION (CLARIFIED JUICE AND SYRUPS)**

Clarified juice or syrup is diluted with distilled water (filtered through a 0.45 micrometre membrane) to  $25 \pm 2^\circ$  Brix, mixed well, and the refractometric Brix of the diluted sample measured. For clear juice specifically, the sample line is flushed and the container rinsed with hot juice immediately before sampling, then cooled to room temperature. Absorbance is read at 900 nm to the nearest 0.001 unit. Turbidity (S) =  $100 \times \text{absorbance}$ .

## **CHAPTER 9: CHALLENGES AND SKILLS GAINED**

### **9.1 TECHNICAL SKILLS GAINED**

The industrial attachment provided practical experience in both **classical** and **instrumental analytical chemistry**.

One of the major skills developed was sample handling. Different factory streams required different sampling methods, preservation procedures, dilution factors and preparation techniques. This demonstrated the importance of representative sampling in obtaining reliable analytical results.

Practical competence was also developed in the use of laboratory instruments. These included the polarimeter, refractometer, brix spindle, drying oven, OxiTop BOD apparatus, COD reactor, pH meter, conductivity meter, spectrophotometer and analytical balance.

The attachment also strengthened titrimetric skills through procedures involving Fehling's solutions, EDTA, hydrochloric acid, silver nitrate, potassium permanganate and other reagents.

Gravimetric techniques were reinforced through moisture, ash, fibre, suspended-solids and water-insoluble-matter determinations.

Another important skill gained was interpretation of analytical results. The attachment demonstrated how individual results such as brix, pol, purity, pH, BOD and COD are related to industrial process performance.

### **9.2 SOFT SKILLS GAINED**

The attachment also contributed to the development of professional skills.

#### **Communication**

Working within the laboratory required clear communication with supervisors, chemists, laboratory assistants and other personnel. Instructions had to be understood correctly before performing analytical procedures.

#### **Teamwork**

Many laboratory activities involved cooperation between laboratory personnel, particularly during sample collection, preparation and routine analysis.

#### **Time management**

The laboratory operated according to routine schedules, with different analyses conducted on particular days and samples collected at specific intervals. This required proper planning and efficient use of time.

#### **Attention to detail**

Analytical chemistry requires careful measurement, observation and recording. Small errors in weighing, pipetting, dilution, titration or reading instruments can affect final results. The attachment therefore strengthened attention to detail and accuracy.

#### **Professional responsibility**

The attachment provided experience in following laboratory procedures, observing safety requirements, maintaining cleanliness and recording analytical information systematically.

### **9.3 CHALLENGES FACED AND HOW THEY WERE ADDRESSED**

The laboratory environment presented several practical challenges associated with industrial analytical work. These included handling samples with different physical properties, working with viscous materials such as syrup and molasses, maintaining accurate measurements and following different preparation procedures for different sample types.

The variety of analyses also required careful attention to the correct procedure, reagent quantities, dilution factors and calculation methods. These challenges were addressed through adherence to laboratory procedures, consultation with experienced laboratory personnel and repeated practice.

### **9.4 LINKING THEORY TO PRACTICE**

The attachment provided a direct connection between analytical chemistry theory and industrial application.

Concepts of instrumental analysis were applied during polarimetry, refractometry, spectrophotometry, pH measurement and conductivity measurements. Titrimetric principles were applied in reducing sugar, hardness, alkalinity, chloride, lime and other analyses.

Gravimetric principles were applied in moisture, ash, fibre, suspended solids and water-insoluble matter determination.

The attachment also demonstrated the application of environmental chemistry through BOD, COD, TSS, conductivity and pH measurements of factory effluent.

Quality assurance and quality control concepts were demonstrated through instrument calibration, standard procedures, sample identification, proper documentation and comparison of results with specified requirements. The report distinguishes QA as the preventive framework supporting reliable laboratory work from QC as the actual testing and verification of samples.

Overall, the attachment transformed theoretical knowledge acquired at JKUAT into practical laboratory competence in an industrial sugar-processing environment.



## CHAPTER 10: CONCLUSION AND RECOMMENDATIONS

### 10.1 INDUSTRIAL ATTACHMENT EXPERIENCE

The industrial attachment at **Nzoia Sugar Company (2025) Limited** provided valuable practical experience in the application of analytical chemistry within a sugar manufacturing environment. The attachment exposed me to the activities of the quality control laboratory and demonstrated how laboratory analysis supports production efficiency, product quality and environmental protection.

During the attachment period, I participated in the collection, preparation and analysis of different samples obtained from various stages of the sugar manufacturing process. This included cane, first expressed juice (FEJ), Mixed Juice (MJ), last expressed juice (LEJ), Clear Juice, Syrup, Massecuite, Molasses, Bagasse, Filter Cake, Sugar and water samples. The laboratory also analysed factory effluent and boiler water to monitor process and environmental conditions.

The attachment provided practical experience in the determination of important sugar-processing parameters such as Brix, Pol, Purity, Reducing Sugars, Moisture, Sulphated Ash, Fibre, Preparation Index, Colour, Turbidity and Suspended Solids. environmental analyses included pH, Conductivity, Total Dissolved Solids (TDS), Total Suspended Solids (TSS), Biochemical Oxygen Demand (BOD<sub>5</sub>) and chemical Oxygen Demand (COD).

The attachment also provided an opportunity to operate and understand the principles of different laboratory instruments, including the polarimeter, Brix Refractometer, Brix Spindle, Drying oven, pH meter, conductivity meter, spectrophotometer, OxiTop bod apparatus, cod reactor and analytical balance.

The results obtained during the attachment demonstrated the importance of continuous laboratory monitoring in industrial production. for example, the effluent results showed progressive reduction in pollutant concentrations through the treatment stages. BOD<sub>5</sub> decreased from 1850 mg/l at the factory influent to 18.5 mg/l at final discharge, while COD decreased from 3920 mg/l to 42.0 mg/l. The final values were within the stated discharge limits in the report.

The attachment therefore provided an important link between theoretical knowledge acquired at JKUAT and its practical application in an industrial setting. Concepts in analytical chemistry, instrumental analysis, titrimetric analysis, gravimetric analysis, quality control and environmental chemistry were applied to actual factory samples.

Overall, the industrial attachment successfully achieved its intended purpose of providing practical exposure to industrial laboratory operations. It improved my technical competence, analytical skills, understanding of sugar manufacturing processes and appreciation of the role of quality control in ensuring consistent product quality and environmental compliance.

## **10.2 CONCLUSION**

The industrial attachment at nzoia sugar company limited was a valuable learning experience that enhanced my understanding of both sugar manufacturing and industrial analytical chemistry.

The attachment demonstrated that the quality control laboratory is an essential component of the sugar manufacturing process. laboratory analysis provides reliable information required for monitoring raw materials, controlling production processes, evaluating final products and assessing the performance of environmental treatment systems.

The determination of brix, pol and purity provided information on the quality and composition of sugar-process streams, while analyses such as Fibre, Preparation Index and Primary Extraction contributed to understanding milling performance. Similarly, Moisture, Ash, Colour and water-insoluble matter were important in evaluating the quality of the final sugar product.

Environmental analyses also demonstrated the importance of laboratory monitoring in protecting the environment. The observed reductions in BOD<sub>5</sub>, COD, TSS, oil and grease and other parameters across the effluent treatment stages showed how analytical measurements can be used to evaluate treatment efficiency.

The attachment further demonstrated the importance of accuracy, precision, proper sampling, correct sample preparation, instrument calibration, adherence to standard procedures and proper recording of results. These practices are essential for generating reliable analytical results upon which industrial decisions can be based.

In conclusion, the attachment strengthened my practical laboratory skills and provided a better understanding of how analytical chemistry contributes to industrial production, quality assurance, quality control and environmental management. It has also provided useful professional experience that will contribute to my future development as an analytical chemist.

## **10.3 RECOMMENDATIONS TO NZOIA SUGAR COMPANY**

Based on the experience gained during the attachment, the following recommendations are proposed:

### **10.3.1 Regular Maintenance and Calibration of Laboratory Equipment**

The company should continue to maintain and calibrate laboratory instruments regularly. Equipment such as polarimeters, refractometers, pH meters, conductivity meters, spectrophotometers and balances require proper maintenance and calibration to ensure reliable and reproducible results.

### **10.3.2 Continued Training of Laboratory Personnel**

Continuous training should be encouraged for laboratory personnel, particularly when new analytical techniques, instruments or procedures are introduced.

Refresher training can also help maintain consistency in sample preparation, analysis, calculations and interpretation of results.

### **10.3.3 Strengthening Sample Identification and Record Keeping**

Proper sample identification, labelling and recording should continue to be emphasized. Accurate records provide traceability and make it easier to investigate unusual results or changes in process performance.

### **10.3.4 Strengthening Laboratory Safety Practices**

Laboratory safety should remain a major priority. Personnel and students should continue to use appropriate personal protective equipment and follow established procedures when handling corrosive chemicals, hot materials, concentrated acids and other hazardous substances.

### **10.3.5 Continued Monitoring of Effluent Treatment**

The company should maintain regular monitoring of the Effluent Treatment Plant. Parameters such as BOD<sub>5</sub>, COD, TSS, pH, conductivity, TDS and oil and grease should continue to be monitored because they provide important information on treatment efficiency and environmental performance.

### **10.3.6 Exposure of Students to Different Laboratory Activities**

Where possible, industrial attachment students should be given opportunities to rotate through different laboratory activities. Exposure to sugar-process analysis, water analysis, effluent analysis and instrumental analysis provides a broader understanding of the role of the laboratory within the factory.

## **10.4 RECOMMENDATIONS TO JKUAT**

The following recommendations are made to improve the industrial attachment programme:

### **10.4.1 Strengthen Practical Laboratory Training**

JKUAT should continue to provide students with sufficient practical exposure to analytical instruments and laboratory techniques before they begin industrial attachment. Familiarity with instruments such as polarimeters, refractometers, spectrophotometers and pH meters would enable students to adapt more quickly to industrial laboratories.

### **10.4.2 Strengthen the Link Between Theory and Industrial Practice**

The university should continue emphasizing the practical application of analytical chemistry concepts taught in class. Units involving instrumental analysis, analytical chemistry, quality control and environmental chemistry are particularly relevant to industrial attachment.

### **10.4.3 Improve Attachment Supervision**

Regular communication between the university supervisor, student and industrial supervisor should be encouraged throughout the attachment period. This would help ensure that students meet the intended learning objectives and receive appropriate guidance.

#### **10.4.4 Encourage Comprehensive Attachment Documentation**

Students should be encouraged to maintain detailed and accurate logbooks throughout the attachment period. Proper documentation makes it easier to prepare the final report and provides evidence of the activities and skills acquired.

#### **10.4.5 Strengthen Industry–University Collaboration**

JKUAT should continue strengthening partnerships with industries such as Nzoia Sugar Company Limited. Stronger collaboration can provide students with more opportunities for industrial exposure, research projects, professional networking and career development.

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## APPENDICES

### APPENDIX A: NZOIA SUGAR COMPANY MASTER PROCESS FLOW DIAGRAM (PFD)

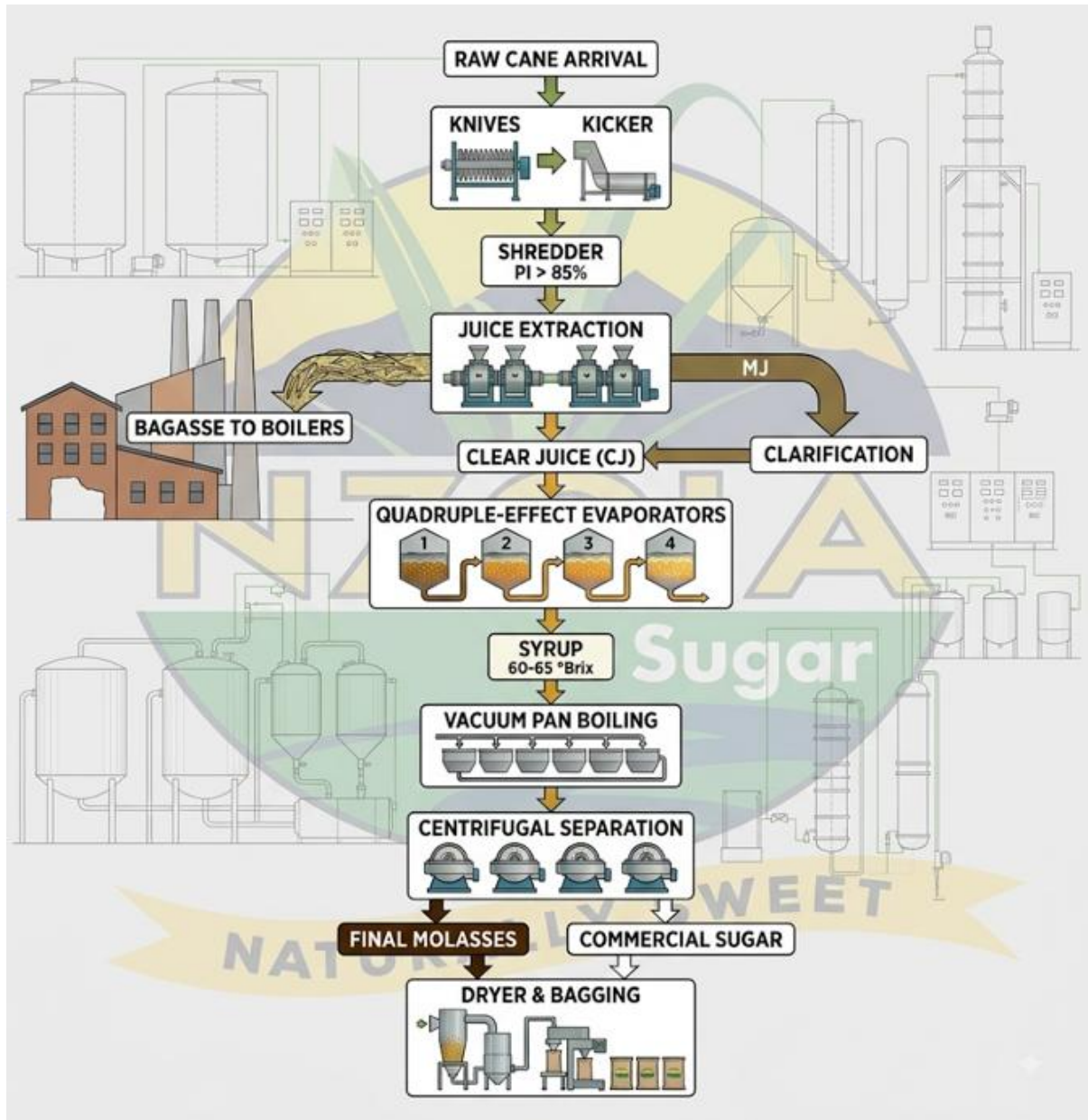


Figure A: process flow diagram (PFD)

## APPENDIX C: MASTER CALCULATIONS & FORMULAE COMPENDIUM

### Summary of Mass Balance and Analytical Formulae used in QC Laboratory

| Analytical Parameter     | Mathematical Expression / Formula                                                                                                         | Key Variables & Definitions                                                                |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Juice Purity (%)         | $\text{Purity (\%)} = \frac{\text{Pol (}^{\circ}\text{Z)}}{\text{Brix (}^{\circ}\text{Bx)}} \times 100$                                   | Pol = Apparent Sucrose %<br><br>Brix = Dissolved Solids %                                  |
| Total Hardness (mg/L)    | $\text{TH} = \frac{V_{\text{EDTA}} \times M_{\text{EDTA}} \times 100,090}{V_{\text{sample}}}$                                             | V = Volume in mL<br><br>M = Molarity of EDTA (0.01 M)                                      |
| Chemical Oxygen Demand   | $\text{COD (mg/L)} = \frac{(V_{\text{blank}} - V_{\text{sample}}) \times N_{\text{FAS}} \times 8000 \times \text{DF}}{V_{\text{sample}}}$ | V = Volume of Titrant (mL)<br><br>N = Normality of FAS Titrant<br><br>DF = Dilution Factor |
| Preparation Index (%)    | $\text{PI (\%)} = \frac{\text{Brix of Unbroken Cells (Tumbler)}}{\text{Total Brix (Disintegrator)}} \times 100$                           | Measures cell disruption efficiency of shredder                                            |
| Hydraulic Retention Time | $\text{HRT (Days)} = \frac{\text{Total Pond Volume (m}^3\text{)}}{\text{Volumetric Flow Rate } Q \text{ (m}^3\text{/day)}}$               | Pond settling design parameter                                                             |

Figure C.1: Formula Summary



## APPENDIX D: PROCESS & LABORATORY PICTORIAL GALLERY



Figure D.1: Ongoing BOD<sub>5</sub> Analysis



Figure D.2: Company overview



Figure D.3: Sampling of Clear Juice

## CONFIRMATION AND SIGNATORIES

I ..... do hereby confirm that I was attached at Nzoia Sugar Company (2025) Ltd for 2 Months from July to August and successfully completed my attachment by acquiring extra skills.

DATE: .....

SIGNATURE.....

### TRAINING OFFICER:

NAMES: .....

COMMENTS:

.....  
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DATE.....

SIGNATURE.....